

WIRESHARK LAB SERIES

Network Traffic Analysis — Beginner to Advanced

⚠ Lab Safety & Ethics Notice

Only capture traffic on a computer and network that you own or are explicitly allowed to test — for example, your own PC on your own home network, or a virtual machine. Never capture on a shared, public, school, workplace, or production network without written permission, and never target another person's device or account.

Activities involving credentials use only (1) throwaway test credentials you make up yourself, (2) the publicly published demo credentials of a service that exists for testing, such as test.rebex.net, or (3) a downloaded sample capture. Never use a real personal login. Unauthorized packet sniffing on networks you do not own or have permission to test is illegal in most jurisdictions.

What You Need Before Starting

- Wireshark installed on a Windows desktop, with administrator rights (required for live capture).
- Your own computer on your own network (home Wi-Fi or Ethernet). A virtual machine or an isolated lab segment is also fine if your instructor provides one, but it is not required.
- Windows Command Prompt (press Start, type `cmd`). Windows 10 and 11 include `curl`, `ping`, `nslookup`, `arp`, and `ftp`. The optional `sftp` step in Activity 5 uses the built-in OpenSSH client.
- Internet access to reach the public test sites: neverssl.com, example.com, httpbin.org, and test.rebex.net.
- For Activity 5: you will pick one of four paths (live HTTP test, live FTP test, downloaded sample capture, or an instructor lab service). The table in Activity 5 helps you choose.

Public test sites can change or go offline without notice. Try each command once before your session. Every activity below includes a backup route, and the Troubleshooting table at the end covers the most common problems.

Getting Plain HTTP Traffic (Read This First)

Wireshark can only read web traffic that is not encrypted. Modern browsers automatically upgrade `http://` addresses to `https://`, and some school firewalls do the same, so a normal browser visit may show only unreadable TLS packets. Use any ONE of the methods below. They all produce the plain HTTP traffic the activities need.

Method 1 — Use curl (recommended)

curl never upgrades to HTTPS on its own, so the output is identical for every student, regardless of browser settings. In Command Prompt run:

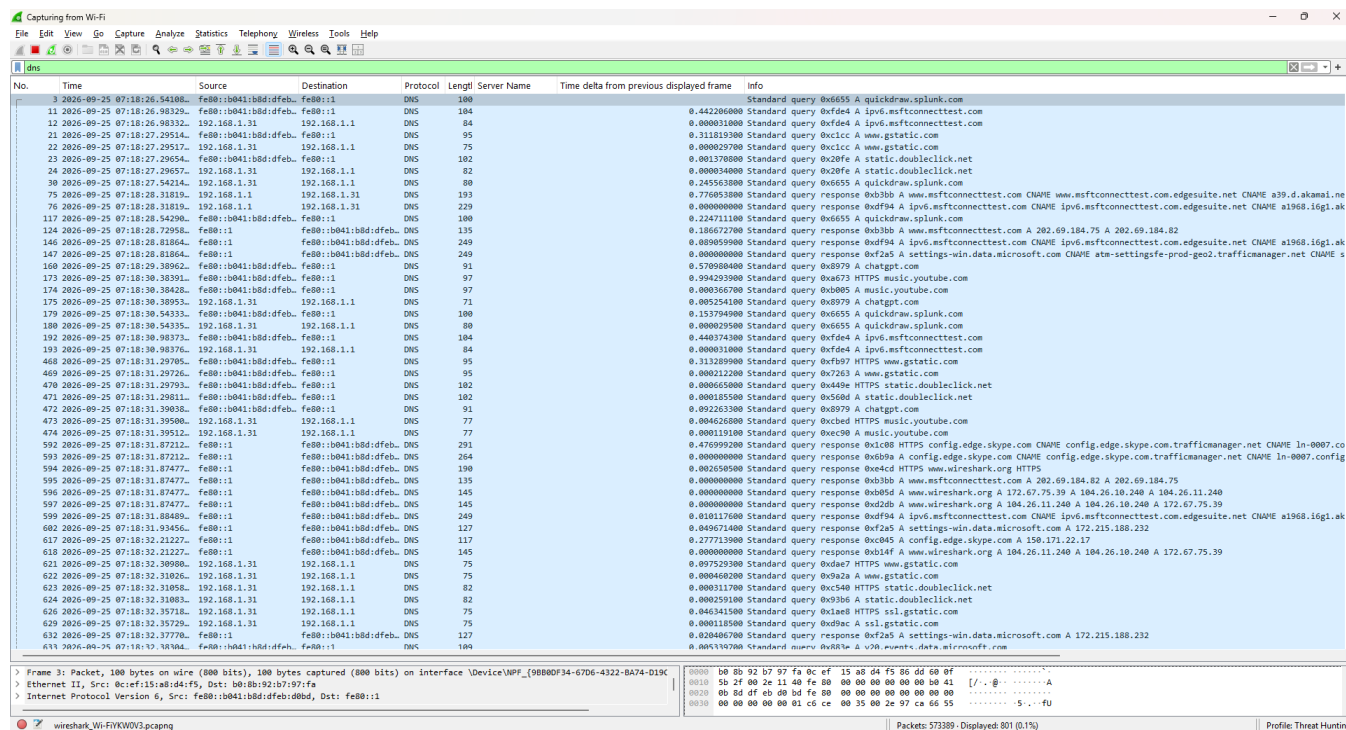
- `curl http://example.com`
- `curl http://neverssl.com`
- If you use PowerShell instead of Command Prompt, type `curl.exe` (PowerShell's plain `curl` is an alias for a different command).

Method 2 — Turn off HTTPS-first mode in your browser

- **Chrome:** Settings → Privacy and security → Security → turn off "Always use secure connections".
- **Edge:** Settings → Privacy, search, and services → turn off "Automatically switch to more secure connections".
- **Firefox:** Settings → Privacy & Security → HTTPS-Only Mode → choose "Don't enable HTTPS-Only Mode".
- Type the full address including `http://` each time. Turn the setting back on when you finish the lab.

Method 3 — Fallbacks

- If `neverssl.com` is blocked or upgraded, use `example.com` for every HTTP-based activity so all students still see identical output.
- If neither loads in a browser, use Method 1 (curl).
- **DNS tip:** browsers may use encrypted DNS (DNS over HTTPS), which hides DNS packets from Wireshark. If you cannot find any DNS packets, run `nslookup example.com` in Command Prompt. It sends classic DNS over UDP port 53 that Wireshark can display.



TCP

Capturing from Wi-Fi									
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help									
tcp									
No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info	
5	2026-09-25 07:18:26.61764	192.168.1.31	20.184.175.23	TCP	66		0.000000000	83927 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
6	2026-09-25 07:18:26.63102	142.250.207.42	192.168.1.31	TCP	54		0.013184400	443 → 58957 [ACK] Seq=1 Ack=1 Win=15595 Len=0	
7	2026-09-25 07:18:26.63102	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=281 Win=15591 Len=0	
8	2026-09-25 07:18:26.63102	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=7081 Win=15595 Len=0	
9	2026-09-25 07:18:26.63102	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=9081 Win=15591 Len=0	
17	2026-09-25 07:18:27.13162	192.168.1.31	104.18.32.47	TLSv1.2	1454		0.506001000	Change Cipher Spec, Application Data, Application Data	
18	2026-09-25 07:18:27.14784	104.177.125.94	192.168.1.31	TLSv1.2	1945		0.012100000	Application Data, Application Data, Application Data	
19	2026-09-25 07:18:27.14784	192.168.1.31	104.177.125.94	TCP	66		0.000040100	[TCP ACKed unseen segment] 59082 → 443 [ACK] Seq=628 Ack=1762 Win=252 Len=0 SLE=1 SRE=992	
20	2026-09-25 07:18:27.25982	192.168.1.31	142.250.197.46	TCP	66		0.112402080	49174 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
25	2026-09-25 07:18:27.33936	142.250.207.42	192.168.1.31	TCP	54		0.079541400	443 → 58957 [ACK] Seq=1 Ack=13607 Win=15587 Len=0	
26	2026-09-25 07:18:27.33936	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=16407 Win=15579 Len=0	
27	2026-09-25 07:18:27.33936	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=19207 Win=15595 Len=0	
28	2026-09-25 07:18:27.33936	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=22807 Win=15591 Len=0	
29	2026-09-25 07:18:27.58032	192.168.1.31	142.250.197.46	TCP	66		0.109012000	56943 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
31	2026-09-25 07:18:27.59738	192.168.1.31	142.250.197.78	TCP	66		0.049009300	58386 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
35	2026-09-25 07:18:27.56708	192.168.1.31	142.250.197.46	TCP	66		0.011313400	52539 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
36	2026-09-25 07:18:27.59032	192.168.1.31	52.168.117.169	TCP	55		0.029623100	56981 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=0	
37	2026-09-25 07:18:27.62057	192.168.1.31	52.168.117.169	TCP	55		0.030245100	58470 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1	
39	2026-09-25 07:18:27.81065	192.168.1.31	142.250.197.78	TCP	66		0.182875000	64552 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
40	2026-09-25 07:18:27.82910	192.168.1.31	104.18.39.21	TLSv1.2	82		0.018453700	Application Data	
41	2026-09-25 07:18:27.86167	142.250.207.42	192.168.1.31	TCP	54		0.032573400	443 → 58957 [ACK] Seq=1 Ack=24807 Win=15591 Len=0	
42	2026-09-25 07:18:27.86167	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=27607 Win=15591 Len=0	
43	2026-09-25 07:18:27.86167	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=30813 Win=15583 Len=0	
44	2026-09-25 07:18:27.86167	142.250.207.42	192.168.1.31	TCP	54		0.000000000	443 → 58957 [ACK] Seq=1 Ack=30579 Win=15600 Len=0	
45	2026-09-25 07:18:27.87251	192.168.1.31	142.250.197.78	TCP	66		0.010840200	53959 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
46	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54		0.013636000	443 → 59028 [FIN, ACK] Seq=1 Ack=1 Win=1047 Len=0	
47	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54		0.000000000	443 → 59028 [ACK] Seq=1 Win=0 Len=0	
48	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54		0.000000000	443 → 55789 [RST] Seq=1 Win=0 Len=0	
49	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54		0.000000000	443 → 55789 [ACK] Seq=1 Ack=1 Win=1047 Len=0	
50	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54		0.000000000	443 → 55789 [RST] Seq=1 Win=0 Len=0	
51	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54		0.000000000	443 → 60499 [ACK] Seq=1 Ack=1 Win=1047 Len=0	
52	2026-09-25 07:18:27.88621	192.168.1.31	142.250.199.238	TCP	54		0.000057900	58210 → 443 [ACK] Seq=1401 Ack=2 Win=255 Len=0	
53	2026-09-25 07:18:27.88624	192.168.1.31	142.250.199.238	TCP	54		0.000030000	55789 → 443 [ACK] Seq=76 Ack=1 Win=255 Len=0	
54	2026-09-25 07:18:27.88633	142.250.199.238	192.168.1.31	TCP	54		0.000064000	443 → 59086 [ACK] Seq=1 Ack=1 Win=1047 Len=0	
55	2026-09-25 07:18:27.88633	142.250.199.238	192.168.1.31	TCP	66		0.000000000	[TCP Retransmission] 49174 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
64	2026-09-25 07:18:28.31689	172.217.26.72	192.168.1.31	TCP	66		0.056235800	443 → 55870 [ACK] Seq=1 Ack=1 Win=1047 Len=0 SLE=4294967223 SRE=1	
65	2026-09-25 07:18:28.31689	142.250.207.42	192.168.1.31	TCP	66		0.000000000	443 → 57630 [ACK] Seq=1 Ack=1 Win=1043 Len=0 SLE=4294967250 SRE=1	
66	2026-09-25 07:18:28.31691	161.49.36.114	192.168.1.31	TCP	54		0.000027100	443 → 57571 [ACK] Seq=1 Ack=1 Win=570 Len=0	
67	2026-09-25 07:18:28.31691	161.49.36.114	192.168.1.31	TCP	54		0.000000000	443 → 57571 [ACK] Seq=1 Ack=1385 Win=593 Len=0	
68	2026-09-25 07:18:28.31694	192.168.1.31	161.49.36.114	TLSv1.2	1438		0.000029100	Ignored Unknown Record	
69	2026-09-25 07:18:28.31694	192.168.1.31	161.49.36.114	TLSv1.2	1438		0.000000000	Ignored Unknown Record	
70	2026-09-25 07:18:28.31696	192.168.1.31	161.49.36.114	TLSv1.2	1438		0.000018100	Ignored Unknown Record	
71	2026-09-25 07:18:28.31696	192.168.1.31	161.49.36.114	TLSv1.2	1438		0.000000000	Ignored Unknown Record	
72	2026-09-25 07:18:28.31819	161.49.36.114	192.168.1.31	TCP	54		0.001232000	443 → 57571 [ACK] Seq=1 Ack=4153 Win=609 Len=0	
73	2026-09-25 07:18:28.31819	161.49.36.114	192.168.1.31	TCP	1262		0.000000000	[TCP Sequence Number] 443 → 57571 [ACK] Seq=4240656246 Ack=4153 Win=609 Len=1200 [TCP RDU reassembled in 80]	
74	2026-09-25 07:18:28.31819	146.75.22.172	192.168.1.31	TCP	66		0.000000000	80 → 65569 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM WS=512	
77	2026-09-25 07:18:28.31819	104.18.39.21	192.168.1.31	TCP	66		0.000000000	443 → 57116 [ACK] Seq=1 Ack=1 Win=6 Len=0 CF=45494967566 CF=1	
Frame 5: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{98800F34-6706-4322-B47A-D19CCE} Ethernet II, Src: \Device\NPF_{98800F34-6706-4322-B47A-D19CCE}, Dst: 08:00:2B:01:00:00 Internet Protocol Version 4, Src: 192.168.1.31, Dst: 20.184.175.23									
wireshark-Wi-Fi-KVM03.pcapng									
Packets: 64268; Displayed: 63590 (99.0%)									
Profile: Threat Hunter									

HTTP

Capturing from Wi-Fi

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http

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
1104	2026-09-25 07:18:36.51589	192.168.1.31	146.75.22.172	HTTP	542			[TCP Retransmission] GET /msdownload/update/v3/static/trusted/en/disallowedcertstl.cab?3aacd52cac28f69 HTTP/1.1
2435	2026-09-25 07:18:45.42392	192.168.1.31	146.75.22.172	HTTP	336			8.910023900 GET /msdownload/update/v3/static/trusted/en/authrootstl.cab?F317465139Fb0082 HTTP/1.1
3421	2026-09-25 07:18:46.24406	146.75.22.172	192.168.1.31	HTTP	256			0.820145800 HTTP/1.1 304 Not Modified
5365	2026-09-25 07:18:46.12367	192.168.1.31	142.250.198.131	HTTP	254			1.879607000 GET /r/r1.cr1 HTTP/1.1
6263	2026-09-25 07:18:46.67681	142.250.198.131	192.168.1.31	HTTP	275			0.553134600 HTTP/1.1 304 Not Modified
21938	2026-09-25 07:19:00.68923	192.168.1.31	142.250.198.131	HTTP	256			12.012428400 GET /r/gsl1.cr1 HTTP/1.1
23137	2026-09-25 07:19:03.65277	142.250.198.131	192.168.1.31	HTTP	276			2.963540300 HTTP/1.1 304 Not Modified
23145	2026-09-25 07:19:03.65849	192.168.1.31	142.250.198.131	HTTP	254			0.005713200 GET /r/r4.cr1 HTTP/1.1
24134	2026-09-25 07:19:05.96727	142.250.198.131	192.168.1.31	HTTP	277			2.380780100 HTTP/1.1 304 Not Modified
26365	2026-09-25 07:19:07.55533	192.168.1.31	146.75.22.172	HTTP	342			1.580858000 GET /msdownload/update/v3/static/trusted/en/disallowedcertstl.cab?eb0ad1145d827f7f HTTP/1.1
27416	2026-09-25 07:19:08.27319	146.75.22.172	192.168.1.31	HTTP	257			0.716608100 HTTP/1.1 304 Not Modified
27418	2026-09-25 07:19:08.27713	192.168.1.31	146.75.22.172	HTTP	336			0.005200900 GET /msdownload/update/v3/static/trusted/en/authrootstl.cab?e157d802c42292eb HTTP/1.1
28628	2026-09-25 07:19:08.67601	146.75.22.172	192.168.1.31	HTTP	256			0.398884600 HTTP/1.1 304 Not Modified
29669	2026-09-25 07:19:10.38773	192.168.1.31	142.250.198.131	HTTP	265			1.717122600 GET /wsl/A40076_Sv7m.cr1 HTTP/1.1
30896	2026-09-25 07:19:15.14824	192.168.1.31	202.69.184.80	HTTP	281			4.769580800 GET /connecttest.txt HTTP/1.1
31188	2026-09-25 07:19:15.16532	142.250.198.131	192.168.1.31	PKIX-CRL	1054			0.017801000 Certificate Revocation List
31787	2026-09-25 07:19:16.12186	202.69.184.80	192.168.1.31	HTTP	241			0.956540900 HTTP/1.1 200 OK (text/plain)
41935	2026-09-25 07:19:37.61252	192.168.1.31	146.75.22.172	HTTP	342			21.490657300 GET /msdownload/update/v3/static/trusted/en/disallowedcertstl.cab?d9a6e6597a7737a HTTP/1.1
42799	2026-09-25 07:19:38.38853	146.75.22.172	192.168.1.31	HTTP	257			0.696012200 HTTP/1.1 304 Not Modified
49610	2026-09-25 07:19:45.79302	192.168.1.31	202.69.184.72	HTTP	201			7.484482100 GET /connecttest.txt HTTP/1.1
50382	2026-09-25 07:19:46.83877	202.69.184.72	192.168.1.31	HTTP	241			1.045750000 HTTP/1.1 200 OK (text/plain)
454488	2026-09-25 07:23:02.39028	192.168.1.31	172.217.26.78	HTTP	466			195.559433300 GET /time/1/current?cpukey=10:f91FXDNG-f3LWXKBVpWp_IP_90r2ZrwUsvVBP-8s8cuphreq=3b0c44298f1c149bf4c8996fb92427ae4ed
455632	2026-09-25 07:23:02.63557	172.217.26.78	192.168.1.31	HTTP/3..	74			0.237366300 HTTP/1.1 200 OK , JSON (application/json)
575493	2026-09-25 07:24:05.34398	192.168.1.31	161.49.25.32	HTTP	281			62.708416200 GET / HTTP/1.1
576385	2026-09-25 07:24:05.58223	161.49.25.32	192.168.1.31	HTTP	317			0.238232900 HTTP/1.1 304 Not Modified

> Frame 1104: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF_{98B0DF34-6706-4322-B474-...}

> Ethernet II, Src: RealtekMediaCtrl, Dst: 08:00:00:00:00:00, Len: 144

> Internet Protocol Version 4, Src: 192.168.1.31, Dst: 146.75.22.172

0000 00 80 92 d7 97 fa 8c ef 15 80 04 f5 00 00 45 00E.....

0008 01 08 f2 a0 00 00 00 00 00 00 00 00 00 00 00P.....

0016 16 ac ff e5 00 50 bf ba 44 15 ab e7 ff 70 50 18H.....P..

0024 00 ff 9b 2a 00 00 47 45 54 20 2f 6d 73 64 6f 77GE T/msdw

dns or tcp or http

Capturing from Wi-Fi

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dns or tcp or http

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
3	2026-09-25 07:18:26.541086	fe8b::b041:b8d:feb...	fe8b::1	DNS	100			Standard query 0x6555 A quickdraw.splunk.com
5	2026-09-25 07:18:26.61764	192.168.1.31	20.184.175.23	TCP	66			0.876752080 63927 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=
6	2026-09-25 07:18:26.63102	142.250.207.42	192.168.1.31	TCP	54			0.813184000 443 → 58957 [ACK] Seq=1 Ack=1 Win=15595 Len=0
7	2026-09-25 07:18:26.63102	142.250.207.42	192.168.1.31	TCP	54			0.800000000 443 → 58957 [ACK] Seq=1 Ack=4281 Win=15591 Len=0
8	2026-09-25 07:18:26.63102	142.250.207.42	192.168.1.31	TCP	54			0.000000000 443 → 58957 [ACK] Seq=1 Ack=7001 Win=15595 Len=0
9	2026-09-25 07:18:26.63102	142.250.207.42	192.168.1.31	TCP	54			0.000000000 443 → 58957 [ACK] Seq=1 Ack=9081 Win=15591 Len=0
11	2026-09-25 07:18:26.98329	fe8b::b041:b8d:feb...	fe8b::1	DNS	104			0.352269600 Standard query 0x6fde4 A ipv6.msftconnecttest.com
12	2026-09-25 07:18:26.98332	192.168.1.31	192.168.1.1	DNS	84			0.000031000 Standard query 0x6fde4 A ipv6.msftconnecttest.com
17	2026-09-25 07:18:27.13162	192.168.1.31	104.18.32.47	TLSv1.2	1454			0.145301300 Change Cipher Spec, Application Data, Application Data
18	2026-09-25 07:18:27.14784	108.177.125.94	192.168.1.31	TLSv1.2	1045			0.016218000 Application Data, Application Data
19	2026-09-25 07:11:27.14784	192.168.1.31	100.177.125.94	TCP	66			0.000040100 [TCP ACK] Unseen segment! 59002 → 443 [ACK] Seq=628 Ack=1762 Win=252 Len=0 SLE=1 SRE=992
20	2026-09-25 07:18:27.29512	192.168.1.31	142.250.197.46	TCP	66			0.811046200 69124 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=
21	2026-09-25 07:18:27.29514	fe8b::b041:b8d:feb...	fe8b::1	DNS	95			0.835317000 Standard query 0xc1cc A www.gstatic.com
22	2026-09-25 07:18:27.29517	192.168.1.31	192.168.1.1	DNS	75			0.800029700 Standard query 0xc1cc A www.gstatic.com
23	2026-09-25 07:18:27.29654	fe8b::b041:b8d:feb...	fe8b::1	DNS	102			0.801370000 Standard query 0x20fe A static.doubleclick.net
24	2026-09-25 07:18:27.29657	192.168.1.31	192.168.1.1	DNS	82			0.800034000 Standard query 0x20fe A static.doubleclick.net
25	2026-09-25 07:18:27.33936	142.250.207.42	192.168.1.31	TCP	54			0.842787200 443 → 58957 [ACK] Seq=1 Ack=13687 Win=15587 Len=0
26	2026-09-25 07:18:27.33936	142.250.207.42	192.168.1.31	TCP	54			0.800000000 443 → 58957 [ACK] Seq=1 Ack=16407 Win=15579 Len=0
27	2026-09-25 07:18:27.33936	142.250.207.42	192.168.1.31	TCP	54			0.800000000 443 → 58957 [ACK] Seq=1 Ack=19207 Win=15595 Len=0
28	2026-09-25 07:18:27.33936	142.250.207.42	192.168.1.31	TCP	54			0.800000000 443 → 58957 [ACK] Seq=1 Ack=22087 Win=15591 Len=0
29	2026-09-25 07:18:27.50838	192.168.1.31	142.250.197.46	TCP	66			0.169016200 56943 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=
30	2026-09-25 07:18:27.54214	192.168.1.31	192.168.1.1	DNS	80			0.837604000 Standard query 0x6555 A quickdraw.splunk.com
31	2026-09-25 07:18:27.55759	192.168.1.31	142.250.197.78	TCP	66			0.815240000 58386 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=
35	2026-09-25 07:18:27.56878	192.168.1.31	142.250.197.46	TCP	66			0.811313400 52539 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=
36	2026-09-25 07:18:27.59832	192.168.1.31	52.168.117.169	TCP	55			0.829623100 56981 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
37	2026-09-25 07:18:27.62857	192.168.1.31	52.168.117.169	TCP	55			0.830245100 58470 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
39	2026-09-25 07:18:27.81069	192.168.1.31	142.250.197.78	TCP	88			0.828477500 64552 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=
40	2026-09-25 07:18:27.82916	192.168.1.31	104.18.39.21	TLSv1.2	82			0.814537000 Application Data
41	2026-09-25 07:18:27.86167	142.250.207.42	192.168.1.31	TCP	54			0.832573400 443 → 58957 [ACK] Seq=1 Ack=24807 Win=15591 Len=0
42	2026-09-25 07:18:27.86167	142.250.207.42	192.168.1.31	TCP	54			0.800000000 443 → 58957 [ACK] Seq=1 Ack=27607 Win=15591 Len=0
43	2026-09-25 07:18:27.86167	142.250.207.42	192.168.1.31	TCP	54			0.800000000 443 → 58957 [ACK] Seq=1 Ack=30811 Win=15583 Len=0
44	2026-09-25 07:18:27.86167	142.250.207.42	192.168.1.31	TCP	54			0.800000000 443 → 58957 [ACK] Seq=1 Ack=30579 Win=15608 Len=0
45	2026-09-25 07:18:27.87251	192.168.1.31	142.250.197.78	TCP	66			0.810840200 53959 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=
46	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54			0.813639000 443 → 50218 [FIN, ACK] Seq=1 Ack=1 Win=1047 Len=0
47	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54			0.800000000 443 → 50218 [RST] Seq=1 Win=0 Len=0
48	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54			0.800000000 443 → 55780 [RST] Seq=1 Win=0 Len=0
49	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54			0.800000000 443 → 55789 [FIN, ACK] Seq=1 Ack=1 Win=1047 Len=0
50	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54			0.800000000 443 → 55789 [RST] Seq=1 Win=0 Len=0
51	2026-09-25 07:18:27.88615	142.250.199.238	192.168.1.31	TCP	54			0.800000000 443 → 60409 [ACK] Seq=1 Ack=1 Win=1047 Len=0
52	2026-09-25 07:18:27.88621	192.168.1.31	142.250.199.238	TCP	54			0.800057900 58210 → 443 [ACK] Seq=1401 Ack=2 Win=255 Len=0
53	2026-09-25 07:18:27.88624	192.168.1.31	142.250.199.238	TCP	54			0.800030900 55789 → 443 [ACK] Seq=76 Ack=2 Win=255 Len=0
54	2026-09-25 07:18:27.88833	142.250.199.238	192.168.1.31	TCP	54			0.800006400 443 → 59886 [ACK] Seq=1 Ack=1 Win=1047 Len=0
55	2026-09-25 07:18:27.92776	192.168.1.31	142.250.197.46	TCP	66			0.821265500 [TCP ACK] Unseen segment! 59117 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=
64	2026-09-25 07:18:28.31689	172.217.26.72	192.168.1.31	TCP	66			0.856326500 443 → 55870 [ACK] Seq=1 Ack=1 Win=1047 Len=0 SLE=4294967223 SRE=1
65	2026-09-25 07:18:28.31689	142.250.207.42	192.168.1.31	TCP	66			0.800000000 443 → 57630 [ACK] Seq=1 Ack=1 Win=1043 Len=0 SLE=4294967258 SRE=1
66	2026-09-25 07:18:28.31691	161.49.36.114	192.168.1.31	TCP	54			0.800027100 443 → 57571 [ACK] Seq=1 Ack=1 Win=570 Len=0
67	2026-09-25 07:18:28.31691	161.49.36.114	192.168.1.31	TCP	54			0.800000000 443 → 57571 [ACK] Seq=1 Ack=1 Win=570 Len=0

> Frame 1104: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF{98080F34-6706-4322-BA74-...}

> Ethernet II, Src: 08:00:15:aa:b4:f5, Dst: 08:00:8b:92:b7:97:fa

> Internet Protocol Version 4, Src: 192.168.1.31, Dst: 146.75.22.172

Info can only be sorted with 10000 or fewer visible rows; increase cache size in Layout preferences

Packets: 1560461 - Displayed: 1469667 (94.2%) Profile: Threat Hunting

- TCP on Frame 6
- DNS on Frame 3
- HTTP on Frame 1104

Activity 2: Capture Filters vs. Display Filters

Level: Beginner

Goal: Learn the difference between filtering before capture (capture filters) and filtering what is already captured (display filters).

Part A — Capture Filter

1. On the Wireshark start screen, click the gear/filter icon next to your interface and enter the capture filter: `host 8.8.8.8`
2. Start the capture, then open Command Prompt and run: `ping 8.8.8.8`
3. Stop the capture and confirm only traffic to/from that host appears. (If your network blocks replies, the outgoing requests are still captured, which is fine.)

The screenshot shows the Wireshark interface with a packet capture filter set to 'host 8.8.8.8'. The packet list displays 8 ICMP Echo (ping) requests and replies. The packet details pane shows the structure of an ICMP Echo request, including the type, code, identifier, and sequence number. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
1	2026-09-28 02:42:28.88798	192.168.1.50	8.8.8.8	ICMP	74			Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 2)
2	2026-09-28 02:42:28.85132	8.8.8.8	192.168.1.50	ICMP	74		0.043339280	Echo (ping) reply id=0x0001, seq=17/4352, ttl=117 (request in 1)
3	2026-09-28 02:42:29.82526	192.168.1.50	8.8.8.8	ICMP	74		0.973943080	Echo (ping) request id=0x0001, seq=18/4668, ttl=128 (reply in 4)
4	2026-09-28 02:42:29.83954	8.8.8.8	192.168.1.50	ICMP	74		0.014281700	Echo (ping) reply id=0x0001, seq=18/4668, ttl=117 (request in 3)
5	2026-09-28 02:42:30.83571	192.168.1.50	8.8.8.8	ICMP	74		0.996166200	Echo (ping) request id=0x0001, seq=19/4864, ttl=128 (reply in 6)
6	2026-09-28 02:42:30.85158	8.8.8.8	192.168.1.50	ICMP	74		0.015872200	Echo (ping) reply id=0x0001, seq=19/4864, ttl=117 (request in 5)
7	2026-09-28 02:42:31.84496	192.168.1.50	8.8.8.8	ICMP	74		0.993362400	Echo (ping) request id=0x0001, seq=20/5128, ttl=128 (reply in 8)
8	2026-09-28 02:42:31.85934	8.8.8.8	192.168.1.50	ICMP	74		0.014376500	Echo (ping) reply id=0x0001, seq=20/5128, ttl=117 (request in 7)

A capture filter with host 8.8.8.8 was set before capture, so only traffic to and from 8.8.8.8 was recorded. The 8 ICMP packets correspond to the 4 pings sent

Part B — Display Filters

1. Start a new, unfiltered capture.
2. Generate plain HTTP traffic: run `curl http://neverssl.com` and `curl http://example.com` (or visit them in a browser with HTTPS-first turned off), plus `nslookup example.com`.
3. Find your default gateway: run `ipconfig` in Command Prompt and copy the "Default Gateway" address.
4. Stop the capture, then apply each of the following display filters one at a time in the filter bar: `dns` | `tcp.port == 80` | `ip.addr == <your default gateway IP>` | `http.request`
5. For each filter, record the resulting packet count shown in the status bar at the bottom of the window.

```
tcp.port == 80
```

ip.addr == 192.168.1.1

Wi-Fi									
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help									
ip.addr == 192.168.1.1									
No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info	
115	2026-09-28 02:48:15.28482.	192.168.1.1	192.168.1.1	DNS	79			Standard query 0x2ace HTTPS accounts.google.com	
116	2026-09-28 02:48:15.28499.	192.168.1.1	192.168.1.1	DNS	79			Standard query 0xb861 A accounts.google.com	
118	2026-09-28 02:48:15.29025.	192.168.1.1	192.168.1.1	DNS	95			Standard query response 0xb861 A accounts.google.com A 108.177.97.84	
120	2026-09-28 02:48:15.30093.	192.168.1.1	192.168.1.1	DNS	129			Standard query response 0x2ace HTTPS accounts.google.com SOA ns1.google.com	
121	2026-09-28 02:48:15.31131.	192.168.1.1	192.168.1.1	DNS	129			Standard query response 0x2ace HTTPS accounts.google.com SOA ns1.google.com	
585	2026-09-28 02:48:30.11822.	192.168.1.1	192.168.1.1	DNS	86			Standard query 0x0763 HTTPS assets-proxy.anthropic.com	14.806509600
586	2026-09-28 02:48:30.11842.	192.168.1.1	192.168.1.1	DNS	86			Standard query 0x4784 A assets-proxy.anthropic.com	0.000196800
587	2026-09-28 02:48:30.13475.	192.168.1.1	192.168.1.1	DNS	139			Standard query response 0x763 HTTPS assets-proxy.anthropic.com HTTPS	0.016329700
598	2026-09-28 02:48:30.13475.	192.168.1.1	192.168.1.1	DNS	182			Standard query response 0x4784 A assets-proxy.anthropic.com A 168.79.104.10	0.000000000
632	2026-09-28 02:48:37.72661.	192.168.1.1	192.168.1.1	DNS	76			Standard query 0x5d94 HTTPS www.facebook.com	7.591861100
633	2026-09-28 02:48:37.72676.	192.168.1.1	192.168.1.1	DNS	76			Standard query 0xe99c A www.facebook.com	0.000156400
634	2026-09-28 02:48:37.73121.	192.168.1.1	192.168.1.1	DNS	92			Standard query response 0xe99c A www.facebook.com A 57.144.104.1	0.005345300
635	2026-09-28 02:48:37.74250.	192.168.1.1	192.168.1.1	DNS	192			Standard query response 0x5d94 HTTPS www.facebook.com CNAME star-mini.c10r.facebook.com HTTPS HTTPS	0.010388000
774	2026-09-28 02:48:43.41763.	192.168.1.1	192.168.1.1	DNS	72			Standard query 0x78a0 A neverssl.com	5.675131900
787	2026-09-28 02:48:46.43492.	192.168.1.1	192.168.1.1	DNS	72			Standard query 0x78a0 A neverssl.com	3.817287600
832	2026-09-28 02:48:50.44376.	192.168.1.1	192.168.1.1	DNS	72			Standard query 0x78a0 A neverssl.com	4.088839000
882	2026-09-28 02:48:51.23350.	192.168.1.1	192.168.1.1	DNS	74			Standard query 0xa786 HTTPS www.google.com	3.769163100
883	2026-09-28 02:48:54.21375.	192.168.1.1	192.168.1.1	DNS	74			Standard query 0xda69 A www.google.com	0.000162900
884	2026-09-28 02:48:54.21905.	192.168.1.1	192.168.1.1	DNS	202			Standard query response 0xda69 A www.google.com A 142.251.157.119 A 142.251.152.119 A 142.251.155.119 A 142.251.151.119 A 14	0.005308600
1049	2026-09-28 02:48:59.74165.	192.168.1.1	192.168.1.1	DNS	74			Standard query response 0xa786 Refused HTTPS www.google.com	5.522594000
1412	2026-09-28 02:49:34.76547.	192.168.1.1	192.168.1.1	DNS	69			Standard query 0x07fc HTTPS Claude.ai	35.022822200
1413	2026-09-28 02:49:34.76567.	192.168.1.1	192.168.1.1	DNS	69			Standard query 0xb410 A Claude.ai	0.000206700
1414	2026-09-28 02:49:34.78110.	192.168.1.1	192.168.1.1	DNS	122			Standard query response 0x07fc HTTPS Claude.ai HTTPS	0.015424500
1415	2026-09-28 02:49:34.78118.	192.168.1.1	192.168.1.1	DNS	85			Standard query response 0xb410 A Claude.ai A 168.79.104.10	0.000000000
1525	2026-09-28 02:49:40.73246.	192.168.1.1	192.168.1.1	DNS	76			Standard query 0xa9a2 HTTPS www.facebook.com	5.351161100
1527	2026-09-28 02:49:40.73263.	192.168.1.1	192.168.1.1	DNS	76			Standard query 0x9475 A www.facebook.com	0.000167300
1530	2026-09-28 02:49:40.76777.	192.168.1.1	192.168.1.1	DNS	92			Standard query response 0x9475 A www.facebook.com A 57.144.228.1	0.035142600
1531	2026-09-28 02:49:40.77968.	192.168.1.1	192.168.1.1	DNS	192			Standard query response 0xa9a2 HTTPS www.facebook.com CNAME star-mini.c10r.facebook.com HTTPS HTTPS	0.011353500

wireshark-Wi-Fi213V3.pcapng

Packets: 1724 - Displayed: 28 (1.6%) - Dropped: 0 (0.0%)

Profile: Threat Hunting

http.request

Wi-Fi									
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help									
http.request									
No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info	
18	2026-09-28 02:48:08.98856.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	
72	2026-09-28 02:48:11.94929.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.960733700
112	2026-09-28 02:48:14.91757.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.968274800
180	2026-09-28 02:48:18.00141.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.083843700
198	2026-09-28 02:48:20.95912.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.957705600
241	2026-09-28 02:48:23.94414.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.985022000
301	2026-09-28 02:48:27.00056.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.856425400
594	2026-09-28 02:48:30.87261.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.872046700
566	2026-09-28 02:48:32.94169.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.869083500
680	2026-09-28 02:48:38.99284.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	6.051147900
739	2026-09-28 02:48:41.95118.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.938354400
782	2026-09-28 02:48:45.82461.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.873427600
796	2026-09-28 02:48:47.99392.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.969318700
843	2026-09-28 02:48:50.68470.	192.168.1.10	34.223.124.45	HTTP	130			GET / HTTP/1.1	2.690775200
851	2026-09-28 02:48:50.96376.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	0.279058900
956	2026-09-28 02:48:57.00415.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	6.040433600
1051	2026-09-28 02:48:59.97377.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.969581700
1057	2026-09-28 02:49:00.25638.	192.168.1.41	239.255.255.250	SSDP	167			H-SEARCH * HTTP/1.1	0.282681300
1075	2026-09-28 02:49:01.94584.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.789461200
1144	2026-09-28 02:49:07.83277.	192.168.1.10	172.66.147.243	HTTP	129			GET / HTTP/1.1	3.986272700
1172	2026-09-28 02:49:08.99386.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	1.961151500
1215	2026-09-28 02:49:12.06291.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.869045200
1237	2026-09-28 02:49:15.02703.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.964120800
1269	2026-09-28 02:49:17.99683.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.969088700
1304	2026-09-28 02:49:24.04051.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	6.043671000
1333	2026-09-28 02:49:27.11822.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.069719100
1361	2026-09-28 02:49:30.00000.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.969051300
1380	2026-09-28 02:49:33.04970.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.969700900
1495	2026-09-28 02:49:39.83075.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	5.980977000
1582	2026-09-28 02:49:42.06104.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.030287900
1598	2026-09-28 02:49:45.02204.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.969093500
1624	2026-09-28 02:49:48.10203.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.080793300
1644	2026-09-28 02:49:51.07237.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.969545200
1675	2026-09-28 02:49:54.13068.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	3.058301500
1712	2026-09-28 02:49:57.12819.	192.168.1.9	239.255.255.250	SSDP	179			H-SEARCH * HTTP/1.1	2.997515600

Request: Boolean

Packets: 1724 - Displayed: 35 (2.0%) - Dropped: 0 (0.0%)

Profile: Threat Hunting

Deliverable

A short table listing each filter used, a one-sentence description of what it shows, and the packet count from the status bar.

Filter	What it shows	Packet count
dns	Only DNS name lookups and responses	42
tcp.port == 80	All TCP traffic to or from port 80, including both curl conversations	24
ip.addr == 192.168.1.1	Every packet sent to or from my default pathway, here all DNS queries and responses	28
http.request	Every packet Wireshark decodes as an HTTP request: my 2 curl GETs plus SSDP M-SEARCH discovery packets from another LAN device	35

Activity 3: Following a TCP Stream & HTTP Analysis

Level: Intermediate

Goal: Understand how a full application-layer conversation appears at the packet level using a known, uniform HTTP target.

1. Start a capture, then request `http://neverssl.com` using ONE of these options: (a) **curl (simplest):** `curl -v http://neverssl.com` — the `-v` flag prints the request and response headers in your window so you can cross-check them against Wireshark; or (b) **browser:** visit `http://neverssl.com` with HTTPS-first turned off. In a browser the page may redirect to a randomized subdomain of `neverssl.com` — this is expected and part of its anti-caching design. `curl` does not follow redirects unless you add `-L`.
2. Stop the capture and apply the display filter: `http.request` or `http.response`
3. Right-click any HTTP packet → Follow → TCP Stream, and read the full reassembled conversation.
4. Record the HTTP method (GET), the Host header, the User-Agent string, and the response status code.
5. Close the stream window. Apply the filter `tcp.flags.syn==1` and locate the three-way handshake (SYN, SYN-ACK, ACK) for that same conversation.
6. Repeat the capture against `http://example.com` (`curl -v http://example.com`) and compare the Host header and response headers between the two sites.

Deliverable

An annotated screenshot of the TCP handshake, plus the Follow TCP Stream text for `neverssl.com` with the four fields labeled, and one sentence comparing it to the `example.com` response.

http.request

Wi-Fi

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

http.request

8x

+

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
1591	2026-09-28 04:50:36.29458...	192.168.1.70	239.255.255.250	SSDP	179			M-SEARCH * HTTP/1.1
1592	2026-09-28 04:50:36.29652...	192.168.1.70	239.255.255.250	SSDP	179			0.001936700 M-SEARCH * HTTP/1.1
1593	2026-09-28 04:50:36.29847...	192.168.1.70	239.255.255.250	SSDP	179			0.001955100 M-SEARCH * HTTP/1.1
2311	2026-09-28 04:51:17.22428...	192.168.1.50	172.66.147.243	HTTP	129			40.925085300 GET / HTTP/1.1
2797	2026-09-28 04:51:51.36538...	192.168.1.50	172.66.147.243	HTTP	129			34.141107900 GET / HTTP/1.1
2950	2026-09-28 04:52:06.91746...	192.168.1.70	239.255.255.250	SSDP	179			15.552076500 M-SEARCH * HTTP/1.1
2951	2026-09-28 04:52:06.91926...	192.168.1.70	239.255.255.250	SSDP	179			0.001902700 M-SEARCH * HTTP/1.1
2952	2026-09-28 04:52:06.92127...	192.168.1.70	239.255.255.250	SSDP	179			0.002007300 M-SEARCH * HTTP/1.1

wireshark_Wi-FiZP10V3.pcapng

Packets: 2991 - Displayed: 8 (0.3%) - Dropped: 0 (0.0%)

Profile: Threat Hunting

http.response

Wi-Fi

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

http.response

8x

+

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
2315	2026-09-28 04:51:17.27854...	172.66.147.243	192.168.1.50	HTTP	59			HTTP/1.1 200 OK (text/html)
2800	2026-09-28 04:51:51.39175...	172.66.147.243	192.168.1.50	HTTP	59			34.121213600 HTTP/1.1 200 OK (text/html)

wireshark_Wi-FiZP10V3.pcapng

Packets: 2991 - Displayed: 2 (0.1%) - Dropped: 0 (0.0%)

Profile: Threat Hunting

1. Four fields from the neverssl.com stream

Field	Value
HTTP method	GET (request line: GET / HTTP/1.1)
Host header	neverssl.com
User-Agent	curl/8.13.0
Response status code	200 OK

2. TCP three-way handshake (neverssl.com (frames 280, 381, 382):

Step	Frame	Direction	Flags
1	280	192.168.1.50 → 34.223.124.45	SYN
2	381	34.223.124.45 → 192.168.1.50	SYN, ACK
3	382	192.168.1.50 → 34.223.124.45	ACK

example.com (frames 2300, 2309, 2310):

Step	Frame	Direction	Flags
1	2300	192.168.1.50 → 172.66.147.243	SYN
2	2309	172.66.147.243 → 192.168.1.50	SYN, ACK
3	2310	192.168.1.50 → 172.66.147.243	ACK

3. One-sentence comparison with example.com

Both sites returned 200 OK to the same GET / request from curl, but neverssl.com is served directly by an Apache server with a fixed Content-Length of 3961 bytes, while example.com (Host: example.com) is served through Cloudflare's CDN using chunked transfer encoding and keep-alive.

Activity 4: ARP, DHCP & Mapping the Local Network

Level: Intermediate

Goal: Use Wireshark to understand Layer 2/3 behavior and inventory devices on your own network.

1. Start a capture. Capture broadly first, then apply the display filter `arp` or `dhcp` after capturing.
2. Force a fresh DHCP handshake using ONE of these options: (a) disable and re-enable your network adapter; or (b) open Command Prompt as Administrator and run `ipconfig /release` followed by `ipconfig /renew`. Expect a brief loss of connectivity. Then filter on `dhcp` (or `bootp` in older Wireshark versions) and identify the Discover, Offer, Request, and Ack packets.
3. Record the DHCP server's IP address and the IP address and lease time assigned to your machine.
4. Open Command Prompt and run: `arp -a`
5. To generate more ARP traffic, ping other devices you own (phone, TV, printer, second computer) using the IP addresses in your router's device list, then run `arp -a` again.
6. In Wireshark, filter on `arp` and correlate the ARP requests/replies with the entries from your `arp -a` output.

7. Hover over each MAC address in the packet details pane — Wireshark auto-resolves the vendor from the MAC prefix.

Many phones use a randomized ("private") MAC address, which shows no vendor name. That is a real privacy feature, not an error — mention it in your write-up if it happens.

Deliverable

A device map (IP, MAC address, vendor) for at least four devices on your network, plus the DHCP lease details you recorded. If your network has fewer than four devices, list every device you found and briefly explain why.

dhcp

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
240	2026-09-28 05:07:29.92316.	192.168.1.50	192.168.1.1	DHCP	342			
250	2026-09-28 05:07:30.01781.	0.0.0.0	255.255.255.255	DHCP	344			0.094654100 DHCP Discover - Transaction ID 0x23a0f1d8
271	2026-09-28 05:07:32.11444.	0.0.0.0	255.255.255.255	DHCP	364			2.096631900 DHCP Request - Transaction ID 0x1040f226
300	2026-09-28 05:07:32.24931.	192.168.1.1	255.255.255.255	DHCP	352			0.134668300 DHCP Offer - Transaction ID 0x1040f226
301	2026-09-28 05:07:32.25004.	0.0.0.0	255.255.255.255	DHCP	370			0.000726200 DHCP Request - Transaction ID 0x1040f226
302	2026-09-28 05:07:32.35216.	192.168.1.1	255.255.255.255	DHCP	352			0.102122000 DHCP ACK - Transaction ID 0x1040f226
1317	2026-09-28 05:07:51.36922.	0.0.0.0	255.255.255.255	DHCP	364			19.017056200 DHCP Request - Transaction ID 0x5ad14203
1495	2026-09-28 05:07:56.69648.	0.0.0.0	255.255.255.255	DHCP	342			5.327261600 DHCP Request - Transaction ID 0xb91f9769
1746	2026-09-28 05:08:00.17029.	192.168.1.50	192.168.1.1	DHCP	358			3.479807100 DHCP Request - Transaction ID 0x3a72c5cf
1767	2026-09-28 05:08:00.43329.	192.168.1.1	192.168.1.50	DHCP	352			0.257002000 DHCP ACK - Transaction ID 0x3a72c5cf

arp

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
8	2026-09-28 05:07:28.13134.	48:c7:3c:ba:f7:d7	ff:ff:ff:ff:ff:ff	ARP	60			Who has 192.168.1.1? Tell 192.168.1.39
85	2026-09-28 05:07:25.93197.	0c:ef:15:ad:cc:03	ff:ff:ff:ff:ff:ff	ARP	42			3.780534900 Who has 192.168.1.68? Tell 192.168.1.27
85	2026-09-28 05:07:25.93225.	48:c7:3c:ba:f7:d7	ff:ff:ff:ff:ff:ff	ARP	60			1.334283900 Who has 192.168.1.1? Tell 192.168.1.39
86	2026-09-28 05:07:25.45497.	d8:c9:09:fd:a0:a1	ff:ff:ff:ff:ff:ff	ARP	60			0.202715500 Who has 192.168.1.1? Tell 192.168.1.79
251	2026-09-28 05:07:36.27886.	b0:80:92:b7:97:fa	ff:ff:ff:ff:ff:ff	ARP	60			4.023882000 Who has 192.168.1.50? Tell 192.168.1.1
272	2026-09-28 05:07:32.11444.	ff:f3:d2:6c:b0:d1	ff:ff:ff:ff:ff:ff	ARP	60			1.015506300 Who has 192.168.1.1? Tell 192.168.1.76
302	2026-09-28 05:07:32.31872.	ff:f3:d2:6c:b0:d1	ff:ff:ff:ff:ff:ff	ARP	60			0.204274100 Who has 192.168.1.76? (ARP Probe)
307	2026-09-28 05:07:32.32442.	48:c7:3c:ba:f7:d7	ff:ff:ff:ff:ff:ff	ARP	60			0.005908000 Who has 192.168.1.1? Tell 192.168.1.39
309	2026-09-28 05:07:32.35123.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	42			0.071042300 Who has 192.168.1.1? Tell 192.168.1.50
322	2026-09-28 05:07:32.40004.	b0:80:92:b7:97:fa	0c:ef:15:ad:d4:f5	ARP	60			0.030519100 192.168.1.1 is at b0:80:92:b7:97:fa
330	2026-09-28 05:07:32.52044.	0c:ef:15:ad:cc:03	ff:ff:ff:ff:ff:ff	ARP	42			0.120402700 Who has 192.168.1.50? Tell 192.168.1.27
331	2026-09-28 05:07:32.52044.	0c:ef:15:ad:d4:f5	0c:ef:15:ad:cc:03	ARP	42			0.000000000 192.168.1.50 is at 0c:ef:15:ad:d4:f5
335	2026-09-28 05:07:32.55043.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	42			0.037970900 Who has 192.168.1.50? Tell 192.168.1.1
336	2026-09-28 05:07:32.55043.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000000000 192.168.1.1 is at b0:80:92:b7:97:fa
337	2026-09-28 05:07:32.55043.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	42			0.000000000 Who has 192.168.1.50? Tell 192.168.1.1
338	2026-09-28 05:07:32.55043.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000000000 192.168.1.1 is at b0:80:92:b7:97:fa
340	2026-09-28 05:07:32.55053.	0c:ef:15:ad:d4:f5	b0:80:92:b7:97:fa	ARP	42			0.000106600 192.168.1.50 is at 0c:ef:15:ad:d4:f5
341	2026-09-28 05:07:32.55054.	0c:ef:15:ad:d4:f5	b0:80:92:b7:97:fa	ARP	42			0.000000000 192.168.1.50 is at 0c:ef:15:ad:d4:f5
342	2026-09-28 05:07:32.62380.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	42			0.064473800 Who has 192.168.1.1? Tell 192.168.1.50
345	2026-09-28 05:07:32.62043.	b0:80:92:b7:97:fa	0c:ef:15:ad:d4:f5	ARP	60			0.027123500 192.168.1.1 is at b0:80:92:b7:97:fa
346	2026-09-28 05:07:32.69077.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.040261300 192.168.1.03 is at 70:08:18:ba:fd:b6
347	2026-09-28 05:07:32.69077.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000000000 192.168.1.50 is at 40:c7:3c:ba:f7:5d
348	2026-09-28 05:07:32.70000.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.001235800 192.168.1.64 is at 24:eb:15:fb:b0:ce
349	2026-09-28 05:07:32.70000.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000000000 192.168.1.69 is at 40:c7:3c:ba:f7:5d
350	2026-09-28 05:07:32.72758.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	42			0.007975700 Who has 192.168.1.50? (ARP Probe)
358	2026-09-28 05:07:33.03955.	a0:29:48:4a:90:d5	ff:ff:ff:ff:ff:ff	ARP	60			0.251970600 Who has 192.168.1.50? Tell 192.168.1.2
359	2026-09-28 05:07:33.03962.	0c:ef:15:ad:d4:f5	a0:29:48:4a:90:d5	ARP	42			0.000078400 192.168.1.50 is at 0c:ef:15:ad:d4:f5
380	2026-09-28 05:07:33.06437.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	42			0.147407800 Who has 192.168.1.50? (ARP Probe)
382	2026-09-28 05:07:33.92089.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.143716300 192.168.1.64 is at 24:eb:15:fb:b0:ce
383	2026-09-28 05:07:33.92089.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000000000 192.168.1.03 is at 70:08:18:ba:fd:b6
384	2026-09-28 05:07:33.93018.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.002090300 192.168.1.09 is at 40:c7:3c:ba:f7:5d
385	2026-09-28 05:07:33.93018.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000000000 192.168.1.50 is at 40:c7:3c:ba:f7:5d
396	2026-09-28 05:07:34.36285.	ff:f3:d2:6c:b0:d1	ff:ff:ff:ff:ff:ff	ARP	60			0.432669400 Who has 192.168.1.76? (ARP Probe)
509	2026-09-28 05:07:34.78023.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	42			0.417407500 Who has 192.168.1.50? (ARP Probe)
618	2026-09-28 05:07:35.28562.	ff:f3:d2:6c:b0:d1	ff:ff:ff:ff:ff:ff	ARP	60			0.005362600 ARP Announcement for 192.168.1.76
714	2026-09-28 05:07:35.77086.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	42			0.492435900 ARP Announcement for 192.168.1.50
838	2026-09-28 05:07:36.96080.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			1.188008400 192.168.1.50 is at 40:c7:3c:ba:f7:5d
839	2026-09-28 05:07:36.96080.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000000000 192.168.1.63 is at 70:08:18:ba:fd:b6
843	2026-09-28 05:07:36.96784.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000087300 192.168.1.69 is at 40:c7:3c:ba:f7:5d
844	2026-09-28 05:07:36.96784.	a0:de:84:05:14:cc	0c:ef:15:ad:d4:f5	ARP	60			0.000000000 192.168.1.64 is at 24:eb:15:fb:b0:ce
851	2026-09-28 05:07:37.00009.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	42			0.030843700 Who has 192.168.1.50? Tell 192.168.1.38
853	2026-09-28 05:07:37.00009.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	60			0.000033000 192.168.1.50 is at 0c:ef:15:ad:d4:f5
871	2026-09-28 05:07:37.23136.	ff:f3:d2:6c:b0:d1	ff:ff:ff:ff:ff:ff	ARP	60			0.238435500 Who has 192.168.1.1? Tell 192.168.1.76
876	2026-09-28 05:07:37.28443.	0c:ef:15:ad:cc:03	0c:ef:15:ad:d4:f5	ARP	42			0.053055700 Who has 192.168.1.27? Tell 192.168.1.50
901	2026-09-28 05:07:37.39044.	0c:ef:15:ad:cc:03	0c:ef:15:ad:d4:f5	ARP	42			0.140626600 192.168.1.27 is at 0c:ef:15:ad:cc:03
917	2026-09-28 05:07:37.78280.	0c:ef:15:ad:d4:f5	ff:ff:ff:ff:ff:ff	ARP	42			0.392447800 ARP Announcement for 192.168.1.50
1188	2026-09-28 05:07:41.77436.	ff:f3:d2:6c:b0:d1	ff:ff:ff:ff:ff:ff	ARP	60			4.404147900 Who has 192.168.1.1? Tell 192.168.1.76

Frame 307: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF{09B80F34-67D6-4322-8A74-0193C6}

Ethernet II, Src: 48:c7:3c:ba:f7:d7, Dst: ff:ff:ff:ff:ff:ff

Address Resolution Protocol (request)

0000 ff ff ff ff ff ff 48 c7 3c ba f7 d7 00 00 00 01
0010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Part 1: DHCP handshake (dhcp filter, Transaction ID 0x1040f226)

Step	Frame	Source → Destination	Packet
1	250	0.0.0.0 → 255.255.255.255	Discover
2	300	192.168.1.1 → 255.255.255.255	Offer
3	301	0.0.0.0 → 255.255.255.255	Request
4	308	192.168.1.1 → 255.255.255.255	ACK

Part 2: DHCP lease details (from frame 308)

Item	Value
DHCP server IP	192.168.1.1
IP address assigned to my machine	192.168.1.50
Lease time	86400 seconds (1 day)
Subnet mask	255.255.255.0

Part 3: Device map

Device	IP	MAC	Vendor
My PC	192.168.1.50	0c:ef:15:a8:d4:f5	TP-Link Systems Inc.
Router / DHCP server	192.168.1.1	b0:8b:92:b7:97:fa	zte corporation
Device 3	192.168.1.2	a8:29:48:4a:9b:d5	TP-Link Systems Inc.
Device 4	192.168.1.27	0c:ef:15:a8:cc:63	TP-Link Systems Inc.
Device 5	192.168.1.38	0c:ef:15:a8:bd:86	TP-Link Systems Inc.
Device 6	192.168.1.39	40:c7:3c:0a:f7:df	Intel Corporate
Device 7	192.168.1.69	40:c7:3c:0d:02:d7	Intel Corporate

Part 4

I released and renewed my DHCP lease with `ipconfig /release` and `/renew`, and captured the four-step exchange: Discover frame 250, Offer 300, Request 301, and ACK (308). The DHCP server is 192.168.1.1, which assigned my PC 192.168.1.50 with a lease time of 86400 seconds (1 day). Using `arp -a` and the arp display filter, I found seven devices on my network and matched their MAC addresses to vendors using Wireshark's Resolved Addresses window. After the renewal, my PC sent ARP probes and an ARP announcement for 192.168.1.50 to check that no other device was using the address. Some devices in the ARP traffic, such as those with MACs `f6:f3:d2:` and `d6:c9:d9:`, showed no vendor name. These use randomized private MAC addresses, a privacy feature common on phones, so a missing vendor is expected and not an error.

Activity 5: Capturing Cleartext Credentials

Level: Advanced**Test Accounts and Public Test Services Only**

This activity uses only (1) made-up credentials sent to a public HTTP testing service, (2) the published demo login of a public FTP test server, (3) a downloaded sample capture, or (4) an instructor-provided lab service. Never point this activity at a real service, a personal account, or any system you do not own or have permission to test.

Goal: See firsthand why unencrypted authentication is insecure, and connect that to the reasoning behind HTTPS, SFTP, and SSH.

Choose Your Path

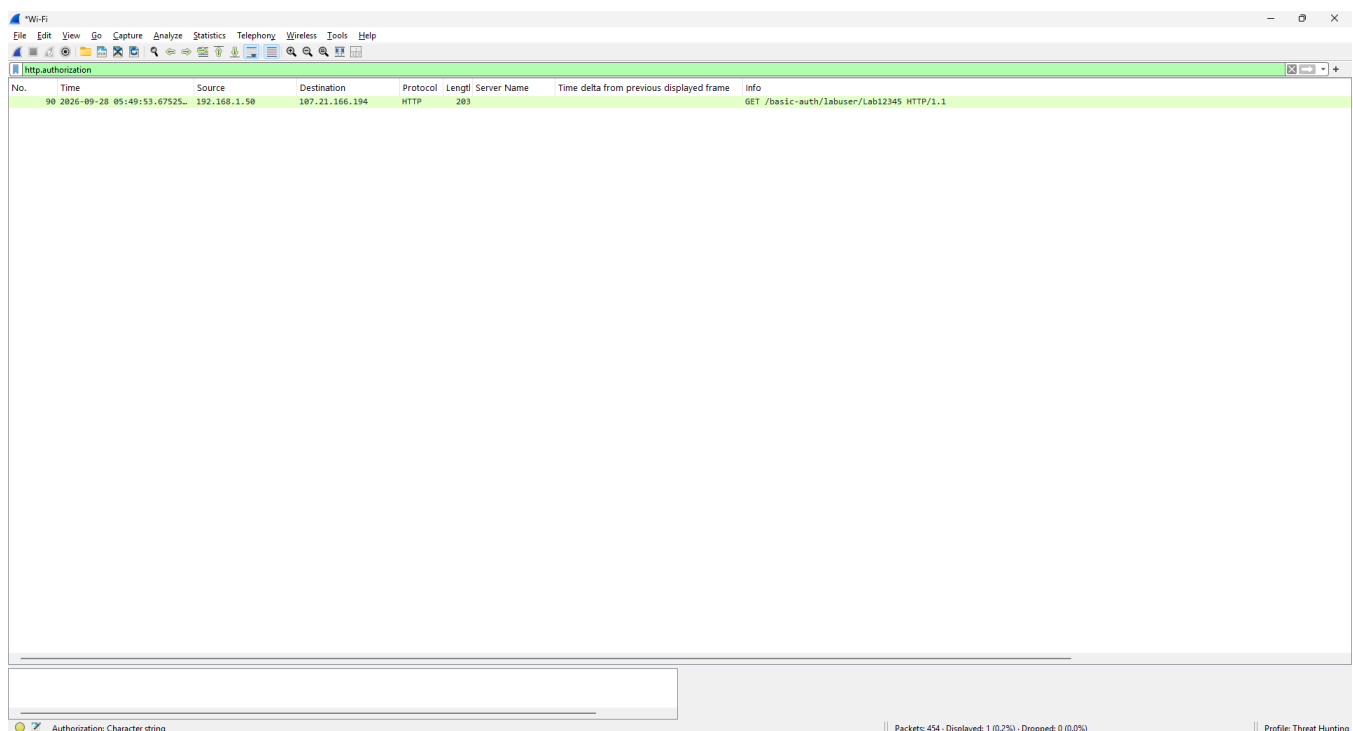
Complete ONE path (A, B, C, or D). Doing a second path is a good way to compare protocols. Paths A and B use live capture; Path C needs no live capture at all.

Path	What you do	Needs	Best when
A — HTTP Basic Auth	Send a made-up login to httpbin.org with curl	Internet	Easiest option; works on almost any network
B — FTP login	Log in to the public test server test.rebex.net with the ftp command	Internet with port 21 open	You want the classic USER/PASS view in Follow TCP Stream
C — Sample capture	Download a .pcap from the Wireshark wiki and analyze it	A download only; no live capture	Safest and most reproducible; also works when test sites are blocked
D — Instructor lab	Use a lab FTP, Telnet, or HTTP Basic Auth service your instructor provides	Lab network	Your instructor has set one up

Path A — HTTP Basic Auth with curl (httpbin.org)

1. Start a Wireshark capture on your active interface.
2. In Command Prompt run: `curl -u labuser:Lab12345 http://httpbin.org/basic-auth/labuser/Lab12345` — you choose the username and password; httpbin accepts whatever you put in the URL. A successful reply looks like `{"authenticated": true, "user": "labuser"}`.
3. Stop the capture, apply the display filter `http.authorization`, and select the GET request.
4. In the packet details pane, expand Hypertext Transfer Protocol → Authorization: Basic. Wireshark decodes the Base64 value inline and shows the plaintext username and password under "Credentials".
5. Right-click the packet → Follow → HTTP Stream to see the complete request, including the Authorization header.

If httpbin.org is down, try <http://httpbingo.org/basic-auth/labuser/Lab12345> instead, or switch to Path B or C.



Wireshark - Packet 90 - Wi-Fi

> Frame 90: Packet, 203 bytes on wire (1624 bits), 203 bytes captured (1624 bits) on interface \Device\NPF_{98880F34-67D6-4322-8A74-D19CC64C5628}, id 0

Ethernet II, Src: 0c:ef:15:a8:d4:f9, Dst: b8:8b:92:b7:97:fa

Internet Protocol Version 4, Src: 192.168.1.50, Dst: 107.21.166.194

Transmission Control Protocol, Src Port: 61652, Dst Port: 80, Seq: 1, Ack: 1, Len: 149

Hypertext Transfer Protocol

GET /basic-auth/labuser/Lab12345 HTTP/1.1\r\n

Host: httpbin.org\r\n

Authorization: Basic bGFlZm91c2pwYXZxhWjWbWQ==\r\n

Credentials: labuser:Lab12345

User-Agent: curl/8.13.0\r\n

Accept: */*\r\n

\r\n

[Response in frame: 100]

[Full request URI: http://httpbin.org/basic-auth/labuser/Lab12345]

0000 b0 80 92 b7 97 fa 0c ef 15 a8 d4 f5 08 00 45 00E-

0010 00 bf f7 ff 40 00 00 06 2e f9 c9 a8 01 32 6b 15 ...@.....2b

0020 a6 c2 f0 d4 00 50 44 56 bb aa 43 95 32 15 50 18PDV..C.2.P

0030 00 ff 75 da 00 00 47 45 54 20 2f 62 61 73 69 63 ...u..GE T /basic

0040 2d 61 75 74 68 2f 6c 61 62 75 73 65 72 2f 4c 61 -auth/labuser/La

0050 62 31 32 33 34 35 20 48 54 54 50 2f 31 2e 31 0d b12345 H TTP/1.1

0060 0a 48 0f 75 74 3a 20 68 74 74 70 62 69 6e 2e 6f ..Host: h ttpbin.o

0070 72 67 0d 0a 41 75 74 68 6f 72 69 7a 61 74 69 6f rg..Auth orizatio

0080 6e 3a 20 42 61 73 69 63 20 62 47 46 69 64 58 4e n: Basic bGFlZm91c2pwYXZxhWjWbWQ=

0090 6c 63 6a 70 4d 59 57 49 70 4d 6a 4d 30 4e 51 3d lcjpwYXZxhWjWbWQ=

00a0 3d 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 63 = User- Agent: c

00b0 75 72 6c 2f 30 2e 31 33 2e 30 0d 0a 41 63 63 65 url/8.13 .0 Acce

00c0 70 74 3a 20 2a 2f 2a 0d 0a 0d 0a pt: */* ..

Packet (203 bytes) Basic Credentials (16 bytes)

No: 90 - Time: 98P - Source: 192.168.1.50 - Destination: 107.21.166.194 - Protocol: HTTP - Length: 203 - Server Name: Time delta from previous displayed frame: 0.00190100 - Info: GET /basic-auth/labuser/Lab12345 HTTP/1.1

Show packet bytes

Layout: Vertical (Stacked)

Close

Help

Wireshark - Follow HTTP Stream (tcp.stream eq 8) - Wi-Fi

GET /basic-auth/labuser/Lab12345 HTTP/1.1

Host: httpbin.org

Authorization: Basic bGFlZm91c2pwYXZxhWjWbWQ==

User-Agent: curl/8.13.0

Accept: */*

HTTP/1.1 200 OK

Date: Mon, 28 Sep 2026 05:49:46 GMT

Content-Type: application/json

Content-Length: 50

Connection: keep-alive

Server: gunicorn/19.9.0

Access-Control-Allow-Origin: *

Access-Control-Allow-Credentials: true

{

"authenticated": true,

"user": "labuser"

}

1 client pkt, 1 server pkt, 1 sum.

Entire conversation (420 bytes)

Show as ASCII

No delta times

Stream 8

Find:

Filter Out This Stream

Print

Save as...

Back

Close

Help

Item	Value
Command used	curl -u labuser:Lab12345 http://httpbin.org/basic-auth/labuser/Lab12345
Display filter	http.authorization
Packet	Frame 90 (GET /basic-auth/labuser/Lab12345 HTTP/1.1), 1 packet displayed

Source → Destination	192.168.1.50 (port 61652) → 107.21.166.194 (httpbin.org, port 80)
Authorization header	Basic bGFidXNlcjpMYWlzMjM0NQ==
Extracted username	labuser
Extracted password	Lab12345
Server response	Frame 100: HTTP/1.1 200 OK, {"authenticated": true, "user": "labuser"}

The username labuser and password Lab12345 were exposed because HTTP Basic Authentication sends credentials as Base64, which is an encoding that anyone can reverse and not encryption, over plain HTTP on port 80. Wireshark decoded them from the Authorization header in a single click, and anyone on the network path could have done the same. HTTPS would have prevented this exposure, because TLS encrypts the entire request, including the Authorization header, so a capture would show only unreadable Application Data. For file-transfer logins, SFTP or SSH would give the same protection that plaintext protocols like FTP and Telnet lack.

Path B — FTP Login (test.rebex.net)

1. Start a Wireshark capture. Optional: use the capture filter `tcp port 21` from Activity 2, or apply the display filter `ftp` after capturing.
2. In Command Prompt run: `ftp test.rebex.net`. When prompted, enter user `demo` and password `password` (these are the published demo credentials for this public test server). A "230 User logged in" reply means it worked.
3. Type `bye` to exit (the Windows ftp client may hang on commands like `dir`; that is fine because the login was already captured), then stop the capture.
4. Apply the display filter `ftp.request.command == "USER"` or `ftp.request.command == "PASS"` to see the two login packets.
5. Right-click a packet → Follow → TCP Stream and locate the USER and PASS lines in plaintext.

Alternative client: if the ftp command is unavailable or hangs, use curl instead: `curl`

`ftp://demo:password@test.rebex.net/` — it logs in and lists the directory. If your school or ISP blocks port 21, use Path A or Path C.

Path C — Downloaded Sample Capture (no live capture)

1. Open the Wireshark Sample Captures page at wiki.wireshark.org/SampleCaptures (the same page is also mirrored in the Wireshark GitLab wiki). Use Ctrl+F to search the page for "telnet" or "ftp".
2. Download a capture that contains a login. Good starting points: `telnet-cooked.pcap` and `telnet-raw.pcap` (Telnet sessions), or `FTpv6-1.cap` and `FTpv6-2.cap` (FTP). Download only from the official Wireshark site or from your instructor.
3. Open the file in Wireshark (File → Open) and apply the display filter `telnet` or `ftp`.
4. Right-click a packet → Follow → TCP Stream and look for a login prompt and the typed username and password. If your chosen file has no login, pick another file.

Other reputable sources: the Netresec public PCAP list (netresec.com/?page=PcapFiles) and the PacketLife capture library. Open unfamiliar captures inside a virtual machine when possible.

Path D — Instructor Lab Service

1. Confirm with your instructor which lab service, test account, and IP address you will use. Then start a Wireshark capture with a display filter matching the service: `tcp.port == 21` for FTP, `tcp.port == 23` for Telnet, or `http for HTTP Basic Auth`.
2. From Command Prompt, connect to the lab service and log in with the provided test credentials (e.g., `ftp <lab-server-ip>`).

- Stop the capture. **FTP/Telnet:** right-click a packet → Follow → TCP Stream and locate the USER and PASS lines in plaintext. **HTTP Basic Auth:** find the Authorization: Basic header in the packet details pane; Wireshark decodes the Base64 value inline.

Extra Credit — See the Fix in Action (optional)

- HTTPS:** run `curl -u labuser:Lab12345 https://httpbin.org/basic-auth/labuser/Lab12345` while capturing. Filter on `tls:` you will see only encrypted "Application Data", and `http.authorization` finds nothing.
- SFTP:** run `sftp demo@test.rebex.net` (type `yes` for the host-key prompt, then the password `password`) while capturing. Filter on `ssh` and try Follow → TCP Stream: the content is unreadable. (If `sftp` is not recognized, see Troubleshooting.)

Deliverable

A screenshot of the extracted username and password (test credentials, demo login, or sample capture only) plus a one-paragraph reflection: which secure protocol (HTTPS, FTPS/SFTP, or SSH) would have prevented this exposure, and why? Extra credit: a screenshot of the encrypted comparison.

HTTPS comparison

tls

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
11	2026-09-28 05:58:04.68215..	172.217.27.101	192.168.1.50	TLSv1.2	127			Application Data
64	2026-09-28 05:58:08.47355..	192.168.1.50	57.144.182.145	TLSv1.2	85			3.991395600 Application Data
66	2026-09-28 05:58:08.87312..	57.144.182.145	192.168.1.50	TLSv1.2	79			0.199565200 Application Data
101	2026-09-28 05:58:10.99104..	192.168.1.50	57.144.182.141	TLSv1.2	86			2.117921900 Application Data
103	2026-09-28 05:58:11.18350..	57.144.182.141	192.168.1.50	TLSv1.2	82			0.192462100 Application Data
114	2026-09-28 05:58:13.39460..	57.144.182.141	192.168.1.50	TLSv1.2	78			2.21105700 Application Data
168	2026-09-28 05:58:16.18038..	192.168.1.50	44.219.150.215	TLSv1.2	516	httpbin.org		2.78572400 Client Hello (SNI=httpbin.org)
171	2026-09-28 05:58:16.40750..	44.219.150.215	192.168.1.50	TLSv1.2	2854			0.227120200 Server Hello
172	2026-09-28 05:58:16.40750..	44.219.150.215	192.168.1.50	TLSv1.2	127			0.000000000 [TCP Previous segment not captured] , Ignored Unknown Record
207	2026-09-28 05:58:16.73581..	192.168.1.50	44.219.150.215	TLSv1.2	188			0.328309100 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
209	2026-09-28 05:58:16.96067..	44.219.150.215	192.168.1.50	TLSv1.2	250			0.224855700 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
212	2026-09-28 05:58:16.96242..	192.168.1.50	44.219.150.215	TLSv1.2	232			0.001750000 Application Data
215	2026-09-28 05:58:17.20252..	192.168.1.50	57.144.182.145	TLSv1.2	1446			0.240106900 Ignored Unknown Record
216	2026-09-28 05:58:17.20252..	192.168.1.50	57.144.182.145	TLSv1.2	1446			0.000000000 Ignored Unknown Record
217	2026-09-28 05:58:17.20252..	192.168.1.50	57.144.182.145	TLSv1.2	1446			0.000000000 Ignored Unknown Record
218	2026-09-28 05:58:17.20252..	192.168.1.50	57.144.182.145	TLSv1.2	1446			0.000000000 Ignored Unknown Record
219	2026-09-28 05:58:17.20252..	192.168.1.50	57.144.182.145	TLSv1.2	1305			0.000000000 Ignored Unknown Record
220	2026-09-28 05:58:17.20250..	192.168.1.50	57.144.182.145	TLSv1.2	83			0.000001100 Application Data
221	2026-09-28 05:58:17.20370..	192.168.1.50	57.144.182.145	TLSv1.2	85			0.000277900 Application Data
222	2026-09-28 05:58:17.20390..	192.168.1.50	57.144.182.145	TLSv1.2	83			0.000120300 Ignored Unknown Record
230	2026-09-28 05:58:17.22707..	192.168.1.50	57.144.182.141	TLSv1.3	829	edge-chat.facebook.com		0.023165000 Client Hello (SNI=edge-chat.facebook.com)
243	2026-09-28 05:58:17.24397..	57.144.182.141	192.168.1.50	TLSv1.3	1446			0.016900100 Server Hello, Change Cipher Spec
244	2026-09-28 05:58:17.24397..	57.144.182.141	192.168.1.50	TLSv1.3	679			0.000000000 Application Data
246	2026-09-28 05:58:17.24433..	192.168.1.50	57.144.182.141	TLSv1.3	118			0.000366100 Change Cipher Spec, Application Data
247	2026-09-28 05:58:17.24465..	192.168.1.50	57.144.182.141	TLSv1.3	1179			0.000313400 Application Data
249	2026-09-28 05:58:17.25995..	57.144.182.141	192.168.1.50	TLSv1.3	239			0.015905100 Application Data
255	2026-09-28 05:58:17.34433..	57.144.182.145	192.168.1.50	TLSv1.2	79			0.000383000 Application Data
257	2026-09-28 05:58:17.36217..	57.144.182.145	192.168.1.50	TLSv1.2	89			0.013837500 Application Data
259	2026-09-28 05:58:17.38035..	57.144.182.145	192.168.1.50	TLSv1.2	98			0.018176800 Application Data
261	2026-09-28 05:58:17.38150..	192.168.1.50	57.144.182.145	TLSv1.2	98			0.001234400 Application Data
262	2026-09-28 05:58:17.39036..	57.144.182.145	192.168.1.50	TLSv1.2	79			0.016779700 Application Data
266	2026-09-28 05:58:17.51901..	44.219.150.215	192.168.1.50	TLSv1.2	312			0.120650800 Application Data
267	2026-09-28 05:58:17.51901..	44.219.150.215	192.168.1.50	TLSv1.2	133			0.000000000 Application Data
269	2026-09-28 05:58:17.52136..	192.168.1.50	44.219.150.215	TLSv1.2	85			0.002341700 Encrypted Alert
276	2026-09-28 05:58:17.79380..	57.144.182.141	192.168.1.50	TLSv1.3	288			0.272440200 Application Data
277	2026-09-28 05:58:17.79494..	192.168.1.50	57.144.182.141	TLSv1.3	438			0.001136400 Application Data
279	2026-09-28 05:58:18.01917..	57.144.182.141	192.168.1.50	TLSv1.3	84			0.224232000 Application Data
290	2026-09-28 05:58:18.03170..	192.168.1.50	57.144.182.145	TLSv1.2	85			0.062527000 Application Data
294	2026-09-28 05:58:19.80864..	57.144.182.145	192.168.1.50	TLSv1.2	79			0.198942600 Application Data

tls and ip.addr == 44.219.150.215

Wi-Fi

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

Filter: tls and ip.addr == 44.219.150.215

127.0.0.1

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
168	2026-09-28 05:58:16.18038...	192.168.1.50	44.219.150.215	TLSv1.2	516	httpbin.org		Client Hello (SNI=httpbin.org)
171	2026-09-28 05:58:16.40750...	44.219.150.215	192.168.1.50	TLSv1.2	2854		0.227126200	Server Hello
172	2026-09-28 05:58:16.40750...	44.219.150.215	192.168.1.50	TLSv1.2	1271		0.000000000	Application Data (TLSv1.2 record not captured) - Ignored Unknown Record
207	2026-09-28 05:58:16.73581...	192.168.1.50	44.219.150.215	TLSv1.2	180		0.320305100	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
209	2026-09-28 05:58:16.96067...	44.219.150.215	192.168.1.50	TLSv1.2	258		0.224058700	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
212	2026-09-28 05:58:16.96242...	192.168.1.50	44.219.150.215	TLSv1.2	232		0.001750000	Application Data
266	2026-09-28 05:58:17.51901...	44.219.150.215	192.168.1.50	TLSv1.2	312		0.556580000	Application Data
267	2026-09-28 05:58:17.51901...	44.219.150.215	192.168.1.50	TLSv1.2	133		0.000000000	Application Data
269	2026-09-28 05:58:17.52136...	192.168.1.50	44.219.150.215	TLSv1.2	85		0.002341700	Encrypted Alert

wireshark_Wi-FiCT7P7V3.pcapng

Packets: 362 · Displayed: 9 (2.5%) · Dropped: 0 (0.0%)

Profile: Threat Hunting

	Path A (HTTP)	Extra credit (HTTPS)
Command	curl -u labuser:Lab12345 http://httpbin.org/...	curl -u labuser:Lab12345 https://httpbin.org/...
Port	80	443
Filter http.authorization	1 packet (frame 90)	0 packets
Credentials in Wireshark	Credentials: labuser:Lab12345	Not visible, only encrypted Application Data
Visible on the wire	Full request, credentials, and response	Only the server name (SNI=httpbin.org), IPs, and packet sizes

SFTP

SSH

Wi-Fi

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

ssh

82 100 +

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
221	2026-09-28 06:05:06.83535..	192.168.1.50	194.108.117.16	SSHv2	87			Client: Protocol (SSH-2.0-OpenSSH_for_Mindows_9.5)
233	2026-09-28 06:05:07.05468..	194.108.117.16	192.168.1.50	SSHv2	81			0.219336200 Server: Protocol (SSH-2.0-RebexSSH_8.0.9742)
234	2026-09-28 06:05:07.05468..	194.108.117.16	192.168.1.50	SSHv2	1198			0.000000000 Server: Key Exchange Init
237	2026-09-28 06:05:07.05529..	192.168.1.50	194.108.117.16	SSHv2	86			0.000605600 Client: Key Exchange Init
256	2026-09-28 06:05:07.30321..	192.168.1.50	194.108.117.16	SSHv2	102			0.247920500 Client: Elliptic Curve Diffie-Hellman Key Exchange Init
257	2026-09-28 06:05:07.52023..	194.108.117.16	192.168.1.50	SSHv2	246			0.217016100 Server: Elliptic Curve Diffie-Hellman Key Exchange Reply
258	2026-09-28 06:05:07.52023..	194.108.117.16	192.168.1.50	SSHv2	70			0.000000000 Server: New Keys
259	2026-09-28 06:05:07.52023..	194.108.117.16	192.168.1.50	SSHv2	434			0.000000000 Server: Encrypted packet (len=300)
366	2026-09-28 06:05:14.00294..	192.168.1.50	194.108.117.16	SSHv2	70			6.482713600 Client: New Keys
369	2026-09-28 06:05:14.27377..	192.168.1.50	194.108.117.16	SSHv2	98			0.270835400 Client: Encrypted packet (len=44)
370	2026-09-28 06:05:14.49166..	194.108.117.16	192.168.1.50	SSHv2	98			0.217908000 Server: Encrypted packet (len=44)
371	2026-09-28 06:05:14.49181..	192.168.1.50	194.108.117.16	SSHv2	114			0.000129000 Client: Encrypted packet (len=60)
372	2026-09-28 06:05:14.73500..	194.108.117.16	192.168.1.50	SSHv2	170			0.243190600 Server: Encrypted packet (len=116)
373	2026-09-28 06:05:14.73500..	194.108.117.16	192.168.1.50	SSHv2	114			0.000000000 Server: Encrypted packet (len=40)
375	2026-09-28 06:05:14.73936..	192.168.1.50	194.108.117.16	SSHv2	138			0.004254000 Client: Encrypted packet (len=84)
376	2026-09-28 06:05:14.95444..	194.108.117.16	192.168.1.50	SSHv2	114			0.215177900 Server: Encrypted packet (len=60)
483	2026-09-28 06:05:20.61587..	192.168.1.50	194.108.117.16	SSHv2	138			5.661436300 Client: Encrypted packet (len=84)
485	2026-09-28 06:05:20.83418..	194.108.117.16	192.168.1.50	SSHv2	114			0.218308300 Server: Encrypted packet (len=60)
486	2026-09-28 06:05:20.83457..	192.168.1.50	194.108.117.16	SSHv2	138			0.000305700 Client: Encrypted packet (len=84)
489	2026-09-28 06:05:21.04091..	194.108.117.16	192.168.1.50	SSHv2	114			0.215343000 Server: Encrypted packet (len=60)
490	2026-09-28 06:05:21.05023..	192.168.1.50	194.108.117.16	SSHv2	138			0.000314000 Client: Encrypted packet (len=84)
495	2026-09-28 06:05:21.26555..	194.108.117.16	192.168.1.50	SSHv2	114			0.215325600 Server: Encrypted packet (len=60)
510	2026-09-28 06:05:21.89615..	192.168.1.50	194.108.117.16	SSHv2	138			1.830592000 Client: Encrypted packet (len=84)
512	2026-09-28 06:05:23.31395..	194.108.117.16	192.168.1.50	SSHv2	114			0.216908200 Server: Encrypted packet (len=60)
661	2026-09-28 06:05:32.91240..	192.168.1.50	194.108.117.16	SSHv2	282			9.599346900 Client: Encrypted packet (len=148)
663	2026-09-28 06:05:33.12758..	194.108.117.16	192.168.1.50	SSHv2	82			0.215094000 Server: Encrypted packet (len=28)
664	2026-09-28 06:05:33.12876..	192.168.1.50	194.108.117.16	SSHv2	106			0.001262700 Client: Encrypted packet (len=52)
668	2026-09-28 06:05:33.34302..	194.108.117.16	192.168.1.50	SSHv2	98			0.214258200 Server: Encrypted packet (len=44)
669	2026-09-28 06:05:33.34365..	192.168.1.50	194.108.117.16	SSHv2	106			0.000630100 Client: Encrypted packet (len=52)
673	2026-09-28 06:05:33.55902..	194.108.117.16	192.168.1.50	SSHv2	126			0.215372900 Server: Encrypted packet (len=72)
675	2026-09-28 06:05:33.55930..	192.168.1.50	194.108.117.16	SSHv2	98			0.000276300 Client: Encrypted packet (len=44)
684	2026-09-28 06:05:33.78851..	194.108.117.16	192.168.1.50	SSHv2	194			0.229210200 Server: Encrypted packet (len=140)
685	2026-09-28 06:05:33.78973..	192.168.1.50	194.108.117.16	SSHv2	98			0.001226100 Client: Encrypted packet (len=44)
686	2026-09-28 06:05:33.78997..	192.168.1.50	194.108.117.16	SSHv2	98			0.000237600 Client: Encrypted packet (len=44)
691	2026-09-28 06:05:34.83462..	194.108.117.16	192.168.1.50	SSHv2	122			0.244653500 Server: Encrypted packet (len=68)
806	2026-09-28 06:05:42.09550..	192.168.1.50	194.108.117.16	SSHv2	90			8.060881100 Client: Encrypted packet (len=36)
813	2026-09-28 06:05:42.31120..	194.108.117.16	192.168.1.50	SSHv2	90			0.215697500 Server: Encrypted packet (len=36)
814	2026-09-28 06:05:42.31141..	192.168.1.50	194.108.117.16	SSHv2	90			0.000206200 Client: Encrypted packet (len=36)
815	2026-09-28 06:05:42.31147..	192.168.1.50	194.108.117.16	SSHv2	114			0.000061000 Client: Encrypted packet (len=60)
823	2026-09-28 06:05:42.52689..	194.108.117.16	192.168.1.50	SSHv2	114			0.215421300 Server: [TCP Out-Of-Order] , Encrypted packet (len=60)

SSH Protocol: Protocol

Packets: 862 - Displayed: 40 (4.6%) - Dropped: 0 (0.0%)

Profile: Threat Hunting

[illegible][illegible]

Item	Value
Command	sftp demo@test.rebex.net (public demo login: demo / password)
Server	test.rebex.net (194.108.117.16), port 22
Display filter	ssh
Packets displayed	40 of 862

Stream	tcp.stream eq 4 (5288 bytes, 21 client packets, 20 server packets)
Client banner	SSH-2.0-OpenSSH for Windows_9.5
Server banner	SSH-2.0-RebexSSH_8.0.9742

Activity 6: Protocol Hierarchy & Anomaly Spotting

Level: Advanced

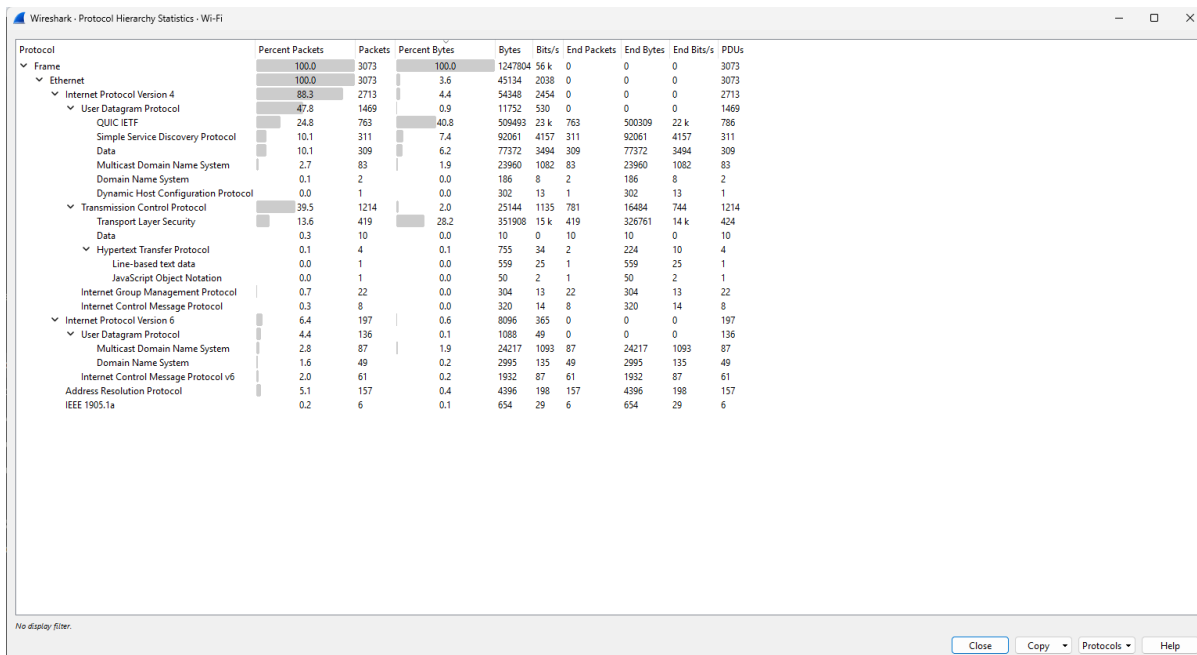
Goal: Use Wireshark's statistics tools to profile a capture's overall traffic mix and flag anything unusual, without relying on packet-by-packet review.

1. Start a capture and generate mixed traffic for about 2 minutes: run `curl http://neverssl.com` and `curl http://example.com`, ping `8.8.8.8`, `nslookup example.com`, and repeat your Activity 5 login command (Path A curl or Path B ftp). **Path C users:** skip this step and open your sample capture (File → Open) instead, then continue with step 2.
2. Stop the capture, then open Statistics → Protocol Hierarchy. Record the top three protocols by percentage of bytes.
3. Open Statistics → Conversations, sort by Bytes, and identify the single conversation (IP pair) that generated the most traffic.
4. Open Statistics → Endpoints and note which single IP address appears in the most conversations.
5. **(Optional)** Apply the display filter `tcp.analysis.flags` to check for retransmissions, duplicate ACKs, or out-of-order packets. A clean home connection often shows none — "none found" is a valid result.
6. Based on steps 2–5, write two sentences describing what a network defender could conclude about this capture just from the statistics views, without opening a single packet.

Deliverable

Screenshots of the Protocol Hierarchy and Conversations windows, plus your two-sentence written analysis.

Protocol Hierarchy



Conversations

Wireshark - Conversations - Wi-Fi

Conversations Settings

☐ Name resolution

☐ Absolute start time

☐ Display raw data

☐ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

☐ Bluetooth

☐ BP7

☐ DCCP

☐ DNP 3.0

☒ Ethernet

☐ FC

☐ FDDI

☐ IEEE 802.11

☐ IEEE 802.15.4

☐ ILNP

☒ IPv4

☐ IPv6

☐ IPX

☐ JXTA

☐ LTP

☐ MPCTCP

☐ NCP

☐ openSAFETY

☐ RSVP

☐ SCTP

☐ SLL

☒ TCP

☐ Token-Ring

☒ UDP

Filter list for specific type

Ethernet - 61 IPv4 - 47 IPv6 - 26 TCP - 21 UDP - 123

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Ref Start	Duration	Bits/s A → B	Bits/s B → A
160.79.104.10	192.168.1.50	675	466 kB	5	262	51 kB	413	415 kB	1.863957300	175.0764910	2325 bits/s	18 kbps
192.168.1.50	192.168.1.50	226	194 kB	42	149	186 kB	77	9 kB	8.182730400	70.202858	21 kbps	969 bits/s
192.168.1.66	239.255.255.250	301	103 kB	7	301	103 kB	0	0 bytes	2.664224600	172.148125	4794 bits/s	0 bits/s
192.168.1.50	57.144.184.1	169	92 kB	1	90	57 kB	79	36 kB	0.741378100	149.793702	3019 bits/s	1897 bits/s
192.168.1.50	57.144.182.141	423	79 kB	4	202	62 kB	221	17 kB	1.809779600	175.133954	2625 bits/s	753 bits/s
192.168.1.50	172.217.27.106	143	67 kB	33	73	42 kB	70	25 kB	60.025014400	81.416162	4161 bits/s	2407 bits/s
192.168.1.50	142.251.8.84	53	35 kB	46	29	23 kB	24	12 kB	152.604002800	0.941406	196 kbps	104 kbps
142.250.198.202	192.168.1.50	167	31 kB	20	86	17 kB	81	14 kB	12.93873100	162.234816	854 bits/s	676 bits/s
192.168.1.4	255.255.255.255	18	16 kB	8	18	16 kB	0	0 bytes	3.181996700	172.553701	758 bits/s	0 bits/s
192.168.1.5	255.255.255.255	17	15 kB	12	17	15 kB	0	0 bytes	5.951380200	162.205323	761 bits/s	0 bits/s
192.168.1.6	255.255.255.255	17	15 kB	15	17	15 kB	0	0 bytes	7.482215900	162.303756	752 bits/s	0 bits/s
192.168.1.50	4.150.223.112	19	14 kB	38	11	3 kB	8	11 kB	76.472819000	1.220115	20 kbps	74 kbps
192.168.1.66	224.0.0.251	12	12 kB	21	12	12 kB	0	0 bytes	13.232430000	160.040078	597 bits/s	0 bits/s
192.168.1.41	224.0.0.251	28	10 kB	6	28	10 kB	0	0 bytes	2.461137300	172.750871	466 bits/s	0 bits/s
192.168.1.50	57.144.182.141	129	9 kB	16	64	4 kB	65	5 kB	7.543190500	169.390991	200 bits/s	218 bits/s
192.168.1.50	172.217.26.67	89	9 kB	29	36	4 kB	53	5 kB	36.146952300	121.87378	240 bits/s	346 bits/s
192.168.1.50	142.251.151.119	52	9 kB	17	27	5 kB	25	3 kB	8.216000200	162.019609	266 bits/s	172 bits/s
192.168.1.50	172.144.229.134	32	6 kB	23	15	1 kB	17	4 kB	21.080696800	154.594657	60 bits/s	222 bits/s
192.168.1.75	224.0.0.251	8	2 kB	36	8	2 kB	0	0 bytes	67.276206000	15.687046	922 bits/s	0 bits/s
192.168.1.50	172.66.147.243	11	2 kB	18	6	411 bytes	5	1 kB	12.474670100	0.077374	42 kbps	118 kbps
192.168.1.42	239.255.255.250	6	1 kB	32	6	1 kB	0	0 bytes	53.456468100	15.975902	537 bits/s	0 bits/s
192.168.1.50	54.175.207.120	9	938 bytes	27	5	431 bytes	4	507 bytes	29.589545500	1.084444	3179 bits/s	3740 bits/s
192.168.1.13	239.255.255.250	4	868 bytes	39	4	868 bytes	0	0 bytes	60.284262000	2.969651	2338 bits/s	0 bits/s
192.168.1.50	142.250.207.37	12	820 bytes	30	6	326 bytes	6	494 bytes	36.560965300	16.064647	162 bits/s	246 bits/s
192.168.1.83	224.0.0.251	8	770 bytes	34	8	770 bytes	0	0 bytes	64.615502400	92.877895	67 bits/s	0 bits/s
192.168.1.50	142.250.199.78	12	773 bytes	24	6	328 bytes	6	445 bytes	24.305829500	149.963082	17 bits/s	23 bits/s
192.168.1.57	224.0.0.251	9	738 bytes	11	9	738 bytes	0	0 bytes	5.428978200	75.573069	78 bits/s	0 bits/s
192.168.1.36	224.0.0.251	7	700 bytes	0	7	700 bytes	0	0 bytes	0.001297900	45.363626	123 bits/s	0 bits/s
192.168.1.50	8.8.8.8	8	592 bytes	19	4	296 bytes	4	296 bytes	12.542145500	3.042171	778 bits/s	778 bits/s
192.168.1.50	108.177.97.188	8	484 bytes	31	4	220 bytes	4	264 bytes	36.765888700	135.176269	13 bits/s	15 bits/s
192.168.1.89	224.0.0.251	5	410 bytes	41	5	410 bytes	0	0 bytes	82.961239700	78.327713	411 bits/s	0 bits/s
192.168.1.48	224.0.0.251	4	386 bytes	14	4	386 bytes	0	0 bytes	71.68431700	145.819707	21 bits/s	0 bits/s
0.0.0.0	255.255.255.255	1	344 bytes	44	1	344 bytes	0	0 bytes	126.77454000	0.000000		
192.168.1.81	224.0.0.251	2	302 bytes	25	2	302 bytes	0	0 bytes	25.914637200	2.041775	1183 bits/s	0 bits/s
192.168.1.50	192.168.1.1	2	270 bytes	26	1	71 bytes	1	199 bytes	29.486191700	0.321902	1764 bits/s	4945 bits/s
192.168.1.20	224.0.0.251	1	240 bytes	45	1	240 bytes	0	0 bytes	146.025645500	0.000000		
192.168.1.41	224.0.0.22	4	216 bytes	40	4	216 bytes	0	0 bytes	82.131118200	30.108610	57 bits/s	0 bits/s
192.168.1.75	224.0.0.22	4	216 bytes	35	4	216 bytes	0	0 bytes	66.663412700	30.412197	56 bits/s	0 bits/s
192.168.1.50	20.184.175.15	3	162 bytes	22	2	108 bytes	1	54 bytes	71.757870700	0.190446	4356 bits/s	2268 bits/s
192.168.1.83	224.0.0.22	3	162 bytes	3	3	162 bytes	0	0 bytes	1.227831200	83.460390	15 bits/s	0 bits/s
192.168.1.57	224.0.0.2	3	138 bytes	10	3	138 bytes	0	0 bytes	5.428978200	74.444178	14 bits/s	0 bits/s
192.168.1.50	172.172.255.216	2	121 bytes	2	1	55 bytes	1	66 bytes	1.047741700	0.340768	1291 bits/s	1549 bits/s
192.168.1.31	224.0.0.251	1	119 bytes	13	1	119 bytes	0	0 bytes	6.647113200	0.000000		
192.168.1.66	224.0.0.22	2	108 bytes	43	2	108 bytes	0	0 bytes	109.056469300	0.615374	1404 bits/s	0 bits/s
192.168.1.81	224.0.0.22	2	108 bytes	28	2	108 bytes	0	0 bytes	30.047767000	0.001208		
192.168.1.27	224.0.0.251	1	85 bytes	37	1	85 bytes	0	0 bytes	73.114693800	0.000000		
192.168.1.36	224.0.0.22	1	54 bytes	9	1	54 bytes	0	0 bytes	3.994467300	0.000000		

Close

Help

Endpoints

Wireshark - Endpoints - Wi-Fi

Endpoint Settings

☐ Name resolution

☐ Display raw data

☐ Hide aggregated

☐ Limit to display filter

Copy

Map

Protocol

☐ Bluetooth

☐ BP7

☐ DCCP

☐ DNP 3.0

☒ Ethernet

☐ FC

☐ FDDI

☐ IEEE 802.11

☐ IEEE 802.15.4

☐ ILNP

☒ IPv4

☐ IPv6

☐ IPX

☐ JXTA

☐ LTP

☐ MPCTCP

☐ NCP

☐ openSAFETY

☐ RSVP

☐ SCTP

☐ SLL

☒ TCP

☐ Token-Ring

☒ UDP

Filter list for specific type

Ethernet - 36 IPv4 - 45 IPv6 - 22 TCP - 36 UDP - 137

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
192.168.1.50	2,344	1.4 kB	1,225	818 kB	1,019	198 kB						
160.79.104.10	675	466 kB	262	51 kB	413	415 kB						
57.144.182.145	423	79 kB	221	17 kB	202	62 kB						
192.168.1.66	315	115 kB	315	115 kB	0	0 bytes						
239.255.255.250	311	105 kB	0	0 bytes	311	105 kB						
57.144.184.141	226	194 kB	77	9 kB	149	186 kB						
57.144.184.1	169	92 kB	79	36 kB	90	57 kB						
142.250.198.202	167	31 kB	86	17 kB	81	14 kB						
172.217.27.106	143	67 kB	70	25 kB	73	42 kB						
57.144.182.141	129	9 kB	65	5 kB	64	4 kB						
172.217.26.67	89	9 kB	53	5 kB	36	4 kB						
224.0.0.251	86	28 kB	0	0 bytes	86	28 kB						
142.251.8.84	53	35 kB	24	12 kB	29	23 kB						
255.255.255.255	53	47 kB	0	0 bytes	53	47 kB						
142.251.151.119	52	9 kB	25	3 kB	27	5 kB						
57.144.229.134	32	6 kB	17	4 kB	15	1 kB						
192.168.1.41	32	10 kB	32	10 kB	0	0 bytes						
4.150.223.112	19	14 kB	8	11 kB	11	3 kB						
192.168.1.4	18	16 kB	18	16 kB	0	0 bytes						
192.168.1.5	17	15 kB	17	15 kB	0	0 bytes						
192.168.1.6	17	15 kB	17	15 kB	0	0 bytes						
224.0.0.22	16	864 bytes	0	0 bytes	16	864 bytes						
142.250.199.78	12	773 bytes	6	445 bytes	6	328 bytes						
142.250.207.37	12	820 bytes	6	494 bytes	6	326 bytes						
192.168.1.57	12	876 bytes	0	0 bytes	0	0 bytes						
192.168.1.75	12	2 kB	12	2 kB	0	0 bytes						
172.66.147.243	11	2 kB	5	1 kB	6	411 bytes						
192.168.1.83	11	940 bytes	11	940 bytes	0	0 bytes						
54.175.207.120	9	938 bytes	4	507 bytes	5	431 bytes						
8.8.8.8	8	592 bytes	4	296 bytes	4	296 bytes						
108.177.97.188	8	484 bytes	4	264 bytes	4	220 bytes						
192.168.1.36	8	754 bytes	8	754 bytes	0	0 bytes						
192.168.1.42	6	1 kB	6	1 kB	0	0 bytes						
192.168.1.89	5	410 bytes	0	0 bytes	0	0 bytes						
192.168.1.15	4	868 bytes	4	868 bytes	0	0 bytes						
192.168.1.48	4	386 bytes	4	386 bytes	0	0 bytes						
192.168.1.81	4	410 bytes	4	410 bytes	0	0 bytes						
20.184.175.15	3	162 bytes	1	54 bytes	2	108 bytes						
224.0.0.2	3	138 bytes	0	0 bytes	3	138 bytes						
172.172.255.216	2	121 bytes	1	66 bytes	1	55 bytes						
192.168.1.1	2	270 bytes	1	199 bytes	1	71 bytes						
0.0.0.0	1	344 bytes	1	344 bytes	0	0 bytes						
192.168.1.20	1	240 bytes	1	240 bytes	0	0 bytes						
192.168.1.27	1	85 bytes	1	85 bytes	0	0 bytes						
192.168.1.31	1	119 bytes	1	119 bytes	0	0 bytes						

Close

Help

Optional anomaly check

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
121	2026-09-28 06:25:26.777077	192.168.1.50	160.79.104.10	TCP	1454			[TCP Retransmission] 51562 + 443 [PSH, ACK] Seq=774 Ack=1 Min=65280 Len=1400 [TCP PDU reassembled in 116]
125	2026-09-28 06:25:27.087739	192.168.1.50	160.79.104.10	TCP	1454			0.310320100 [TCP Retransmission] 51562 + 443 [ACK] Seq=1 Ack=1 Min=65280 Len=1400
132	2026-09-28 06:25:27.155575	192.168.1.50	57.144.182.141	TCP	86			0.078121000 [TCP Retransmission] 51843 + 443 [PSH, ACK] Seq=1 Ack=1 Min=251 Len=32
137	2026-09-28 06:25:27.475556	192.168.1.50	57.144.182.141	TCP	86			0.310018100 [TCP Retransmission] 51843 + 443 [PSH, ACK] Seq=1 Ack=1 Min=251 Len=32
145	2026-09-28 06:25:27.536377	160.79.104.10	192.168.1.50	TCP	66			0.060955000 [TCP Dup ACK 144#1] 443 + 51562 [ACK] Seq=1 Ack=2174 Min=131072 Len=0 SLE=774 SRE=2174
147	2026-09-28 06:25:27.536444	192.168.1.50	160.79.104.10	TCP	827			0.000064000 [TCP Spurious Retransmission] 51562 + 443 [PSH, ACK] Seq=1401 Ack=1 Min=65280 Len=773 [TCP PDU reassembled in 116]
150	2026-09-28 06:25:27.537511	160.79.104.10	192.168.1.50	TCP	66			0.001065000 [TCP Dup ACK 144#2] 443 + 51562 [ACK] Seq=1 Ack=2174 Min=131072 Len=0 SLE=1 SRE=1401
168	2026-09-28 06:25:27.832821	192.168.1.50	160.79.104.10	TCP	118			0.295319200 [TCP Retransmission] 51562 + 443 [PSH, ACK] Seq=2174 Ack=1376 Min=64000 Len=64
171	2026-09-28 06:25:28.083871	57.144.182.141	192.168.1.50	TCP	82			0.251040800 [TCP Spurious Retransmission] 443 + 51843 [PSH, ACK] Seq=1 Ack=33 Min=2031 Len=28
172	2026-09-28 06:25:28.083992	192.168.1.50	57.144.182.141	TCP	66			0.000043000 [TCP Dup ACK 167#1] 51843 + 443 [ACK] Seq=33 Ack=29 Min=251 Len=0 SLE=1 SRE=29
173	2026-09-28 06:25:28.143111	192.168.1.50	160.79.104.10	TCP	118			0.061213500 [TCP Retransmission] 51562 + 443 [PSH, ACK] Seq=2238 Ack=1888 Min=62000 Len=64
174	2026-09-28 06:25:28.318341	57.144.182.141	192.168.1.50	TCP	82			0.165205800 [TCP Spurious Retransmission] 443 + 51843 [PSH, ACK] Seq=1 Ack=33 Min=2031 Len=28
175	2026-09-28 06:25:28.318361	192.168.1.50	57.144.182.141	TCP	66			0.000020400 [TCP Dup ACK 167#2] 51843 + 443 [ACK] Seq=33 Ack=29 Min=251 Len=0 SLE=1 SRE=29
177	2026-09-28 06:25:28.442324	57.144.182.141	192.168.1.50	TCP	54			0.131981300 [TCP Dup ACK 168#1] 443 + 51843 [ACK] Seq=29 Ack=33 Min=2031 Len=0
183	2026-09-28 06:25:28.553332	160.79.104.10	192.168.1.50	TCP	54			0.112799600 [TCP Dup ACK 182#1] 51602 + 443 [ACK] Seq=2238 Ack=1888 Min=62000 Len=0
184	2026-09-28 06:25:28.567775	57.144.182.141	192.168.1.50	TCP	54			0.012435200 [TCP Dup ACK 168#2] 443 + 51843 [ACK] Seq=29 Ack=33 Min=2031 Len=0
197	2026-09-28 06:25:28.940881	57.144.182.141	192.168.1.50	TCP	82			0.373055000 [TCP Spurious Retransmission] 443 + 51843 [PSH, ACK] Seq=1 Ack=33 Min=2031 Len=28
198	2026-09-28 06:25:28.940903	192.168.1.50	57.144.182.141	TCP	66			0.000023300 [TCP Dup ACK 167#3] 51843 + 443 [ACK] Seq=33 Ack=29 Min=251 Len=0 SLE=1 SRE=29
211	2026-09-28 06:25:29.064668	160.79.104.10	192.168.1.50	TCP	66			0.923607500 [TCP Dup ACK 181#1] 443 + 51562 [ACK] Seq=1888 Ack=2238 Min=131072 Len=0 SLE=2174 SRE=2238
217	2026-09-28 06:25:30.482222	160.79.104.10	192.168.1.50	TCP	66			0.563541100 [TCP Dup ACK 181#2] 443 + 51562 [ACK] Seq=1888 Ack=2238 Min=131072 Len=0 SLE=2174 SRE=2238
474	2026-09-28 06:25:41.540431	57.144.182.145	192.168.1.50	TCP	90			11.112212100 [TCP Retransmission] 443 + 54020 [PSH, ACK] Seq=144 Ack=103 Min=2029 Len=36
617	2026-09-28 06:25:51.053577	192.168.1.50	57.144.182.145	TCP	122			9.494140200 [TCP Retransmission] 54020 + 443 [PSH, ACK] Seq=273 Ack=1330 Min=252 Len=88
621	2026-09-28 06:25:51.118881	57.144.182.141	192.168.1.50	TCP	83			0.084316000 [TCP Retransmission] 443 + 51043 [PSH, ACK] Seq=204 Ack=45 Min=2031 Len=29
623	2026-09-28 06:25:51.143906	57.144.182.145	192.168.1.50	TCP	54			0.024182400 [TCP Dup ACK 620#1] 443 + 54020 [ACK] Seq=1320 Ack=341 Min=2029 Len=0
666	2026-09-28 06:25:55.111841	192.168.1.50	57.144.182.145	TCP	88			3.968775000 [TCP Retransmission] 54020 + 443 [PSH, ACK] Seq=341 Ack=1394 Min=252 Len=34
670	2026-09-28 06:25:55.224336	57.144.182.145	192.168.1.50	TCP	54			0.112525500 [TCP Dup ACK 667#1] 443 + 54020 [ACK] Seq=1394 Ack=375 Min=2029 Len=0
676	2026-09-28 06:26:12.792931	57.144.182.145	192.168.1.50	TCP	90			17.568566000 [TCP Spurious Retransmission] 443 + 50467 [PSH, ACK] Seq=204 Ack=15302 Min=2450 Len=44
677	2026-09-28 06:26:12.792977	192.168.1.50	57.144.182.145	TCP	66			0.000037200 [TCP Dup ACK 673#1] 50467 + 443 [ACK] Seq=15138 Ack=253 Min=252 Len=0 SLE=209 SRE=253
914	2026-09-28 06:26:15.440481	192.168.1.50	160.79.104.10	TCP	55			2.647435200 [TCP Keep-Alive] 51562 + 443 [ACK] Seq=2237 Ack=1888 Min=65280 Len=1
915	2026-09-28 06:26:15.466664	160.79.104.10	192.168.1.50	TCP	66			0.026232600 [TCP Keep-Alive ACK] 443 + 51562 [ACK] Seq=1888 Ack=2238 Min=131072 Len=0 SLE=2237 SRE=2238
920	2026-09-28 06:26:16.561368	192.168.1.50	160.79.104.10	TCP	55			1.097939200 [TCP Keep-Alive] 50200 + 443 [ACK] Seq=63618 Ack=3620 Min=55640 Len=1
921	2026-09-28 06:26:16.581227	160.79.104.10	192.168.1.50	TCP	66			0.017595900 [TCP Keep-Alive ACK] 443 + 50200 [ACK] Seq=3620 Ack=63611 Min=55640 Len=0 SLE=63610 SRE=63611
1103	2026-09-28 06:26:23.471336	57.144.182.145	192.168.1.50	TCP	79			6.800091800 [TCP Spurious Retransmission] 443 + 56674 [PSH, ACK] Seq=151 Ack=218 Min=2031 Len=25
1104	2026-09-28 06:26:23.471432	192.168.1.50	57.144.182.145	TCP	66			0.000005400 [TCP Dup ACK 109#1] 56674 + 443 [ACK] Seq=218 Ack=176 Min=509 Len=0 SLE=151 SRE=176
1108	2026-09-28 06:26:23.528661	192.168.1.50	57.144.182.145	TCP	83			0.287273700 [TCP Retransmission] 50467 + 443 [PSH, ACK] Seq=15138 Ack=253 Min=252 Len=29
1113	2026-09-28 06:26:24.000466	57.144.182.145	192.168.1.50	TCP	79			0.241804100 [TCP Spurious Retransmission] 443 + 56674 [PSH, ACK] Seq=151 Ack=218 Min=2031 Len=25
1114	2026-09-28 06:26:24.000511	192.168.1.50	57.144.182.145	TCP	66			0.000047300 [TCP Dup ACK 109#2] 56674 + 443 [ACK] Seq=218 Ack=176 Min=509 Len=0 SLE=151 SRE=176
1116	2026-09-28 06:26:24.079625	192.168.1.50	57.144.182.145	TCP	83			0.009739100 [TCP Retransmission] 50467 + 443 [PSH, ACK] Seq=15138 Ack=253 Min=252 Len=29
1130	2026-09-28 06:26:24.214861	57.144.182.145	192.168.1.50	TCP	54			0.144818100 [TCP Dup ACK 1129#1] 443 + 50467 [ACK] Seq=253 Ack=15167 Min=2450 Len=0
1131	2026-09-28 06:26:24.214861	57.144.182.145	192.168.1.50	TCP	54			0.000000000 [TCP Dup ACK 1129#2] 443 + 50467 [ACK] Seq=253 Ack=15167 Min=2450 Len=0
1136	2026-09-28 06:26:27.364544	57.144.184.1	192.168.1.50	TCP	673			3.149688000 [TCP Spurious Retransmission] 443 + 51145 [PSH, ACK] Seq=1393 Ack=2200 Min=521472 Len=619 [TCP PDU reassembled in 118]
1197	2026-09-28 06:26:27.364595	192.168.1.50	57.144.184.1	TCP	66			0.000004000 [TCP Dup ACK 1130#1] 51145 + 443 [ACK] Seq=6096 Ack=2012 Min=65280 Len=0 SLE=1393 SRE=2012
1200	2026-09-28 06:26:27.545454	57.144.184.1	192.168.1.50	TLSv1.3	89			0.100044000 [TCP Previous segment not captured] Application Data
1201	2026-09-28 06:26:27.545454	57.144.184.1	192.168.1.50	TCP	134			0.000000000 [TCP Retransmission] 443 + 51145 [PSH, ACK] Seq=2191 Ack=2356 Min=521472 Len=80
1202	2026-09-28 06:26:27.545454	57.144.184.1	192.168.1.50	TCP	85			0.000000000 [TCP Retransmission] 443 + 51145 [PSH, ACK] Seq=2271 Ack=2356 Min=521472 Len=31
1208	2026-09-28 06:26:27.686451	57.144.184.1	192.168.1.50	TCP	89			0.009091800 [TCP Spurious Retransmission] 443 + 51145 [PSH, ACK] Seq=2382 Ack=5096 Min=518012 Len=35
1209	2026-09-28 06:26:27.686451	192.168.1.50	57.144.184.1	TCP	66			0.000004300 [TCP Dup ACK 1208#1] 51145 + 443 [ACK] Seq=5102 Ack=1336 Min=60241 Min=52530 SRE=2337

> Frame 121: Packet, 1454 bytes on wire (11632 bits), 1454 bytes captured (11632 bits) on interface \Device\NPF_{98B00F34-67D6-4322-E...}

> Ethernet II, Src: Ocieff:15:ad:d4:f5, Dst: b0:0b:92:1b:97:fa

> Internet Protocol Version 4, Src: 192.168.1.50, Dst: 160.79.104.10

0000 b0 0b 92 1b 97 fa bc ef 15 a8 d4 f5 00 00 45 00@...2...Q

0010 05 a0 d1 11 40 00 00 06 5a 12 c0 a0 01 32 a0 4f h...j...f...c...P

0020 68 0a c9 6a 01 b0 9d c3 11 2f a8 fc 63 75 50 18 h...j...f...c...P

0030 00 ff 06 30 00 00 30 29 a3 16 59 d2 35 7b 47 a3 --0-0-0-Y-5[G

TCP Analysis Flags Label

Packets: 3073 - Displayed: 138 (4.5%) - Dropped: 0 (0.0%)

Profile: Threat-Hunting

The Protocol Hierarchy shows the capture is dominated by encrypted traffic, QUIC (40.8% of bytes) and TLS (28.2%), with only 4 plaintext HTTP packets (0.1%), and Conversations shows the busiest flow was 192.168.1.50 ↔ 160.79.104.10 at 466 kB, mostly upload from my PC (415 kB vs. 51 kB). From these views alone, a defender could conclude that nearly all content is protected and that the small amount of unencrypted HTTP is the part worth inspecting for exposed credentials, while the one-sided heavy flow and the 138 anomaly-flagged packets (4.5%) deserve a follow-up to confirm they are benign.

Capstone Activity: Independent Traffic Investigation

Level: Advanced

No step-by-step instructions are provided for this activity. Use everything from Activities 1–6 as your toolkit.

Scenario

You have been asked to investigate a machine for unusual behavior. Choose one of the following data sources:

- (a) **Your own live capture (about 5 minutes):** capture your own traffic while you browse or use curl against <http://neverssl.com> and <http://example.com>, transfer a file (for example `curl -u demo:password -o readme.txt ftp://test.rebex.net/pub/example/readme.txt`), and repeat your Activity 5 login.
- (b) **A .pcapng file provided by your instructor**, if they supply one.
- (c) **A sample capture from the Wireshark wiki** (wiki.wireshark.org/SampleCaptures) that includes a login, such as a Telnet or FTP session from Activity 5, Path C.

Your Task

Produce a one-page incident-style report that includes:

- Protocols present:** a summary of the protocols and services in the capture (Statistics → Protocol Hierarchy can help).
- Plaintext exposure:** any plaintext credentials or sensitive data you found, and which packets or streams they came from.

3. **Key conversation:** one significant conversation with its timestamp, source/destination IPs, and ports.
4. **Your own filters and recommendations:** at least two display filters you constructed yourself (not copied from earlier activities), each with one sentence explaining your reasoning, followed by two or three recommendations describing what protocol, configuration, or network segmentation changes would reduce the exposure you found.

Constraints: you decide which Wireshark features to use — Follow Stream, Statistics menus, Export Objects, Conversations, I/O graphs, or anything else covered in this lab series. Grading is based on the quality of your findings and the display filters you build, not on following a fixed sequence.

Deliverable

A written one-page incident-style report covering the four required parts above, submitted with supporting screenshots.

Incident Report: Cleartext Credential Exposure

1. Protocols Present

The capture is mostly encrypted by volume: **TLS 38.6%** and **QUIC 34.5%** of bytes, with unclassified UDP data at 17.2%. Cleartext application traffic is small but sensitive: **4 HTTP packets** and **22 FTP packets** (0.1% and 0.6% of packets). Local discovery traffic (SSDP, mDNS, NetBIOS, DHCP, ARP) also appears on the LAN.

2. Plaintext Exposure

- FTP login (TCP stream 13):** USER demo (frame 557) and PASS password (frame 561) sent to 194.108.117.16:21, confirmed by 230 User 'demo' logged in (frame 562). The file readme.txt (379 bytes) was also transferred unencrypted (frame 594).
- HTTP Basic Auth (TCP stream 17):** frame 678, GET /basic-auth/labuser/Lab12345 to httpbin.org (34.200.179.76:80). Wireshark decoded labuser:Lab12345 from the Authorization header; the credentials also appear in the URL path. Server reply: frame 701, 200 OK.

All credentials are published demo or made-up test values used on my own network.

3. Key Conversation

FTP control connection: **192.168.1.50:59734 → 194.108.117.16:21** (test.rebex.net), 06:37:26.97 to 06:37:30.22 on 2026-09-28 (about 3.25 s, TCP stream 13). The file transfer used a second connection to port 1041 (Extended Passive Mode). The entire login, directory navigation, and download are readable in Follow TCP Stream, and the session closed cleanly with no resets.

4. My Filters and Recommendations

Display filter	Why I built it (edit in your own words)	Result
frame contains "labuser"	<i>Searches every byte of every packet, so it finds leaked strings in any protocol or port that a protocol-specific filter could miss.</i>	2 packets (678, 701)
ip.src == 192.168.1.50 && (tcp.port == 80 tcp.port == 21)	<i>Isolates everything my PC sent over unencrypted ports, which shows every cleartext destination at once.</i>	28 packets, 3 servers
tcp.flags.reset == 1	<i>Checks for forcibly terminated connections, which could point to interference or a failed session.</i>	0 packets

Wireshark - Protocol Hierarchy Statistics - capstone.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
Frame	100.0	3945	100.0	2764553	354 k	0	0	0	3945
Ethernet	100.0	3945	2.1	58234	7458	0	0	0	3945
Internet Protocol Version 6	5.3	211	0.3	8816	1129	0	0	0	211
User Datagram Protocol	3.4	134	0.0	1072	137	0	0	0	134
Multicast Domain Name System	2.2	86	0.8	21288	2726	86	21288	2726	86
Link-local Multicast Name Resolution	0.2	7	0.0	231	29	7	231	29	7
Domain Name System	0.4	17	0.0	1056	135	17	1056	135	17
DHCPv6	0.6	24	0.1	2280	292	24	2280	292	24
Internet Control Message Protocol v6	2.0	77	0.1	2352	301	77	2352	301	77
Internet Protocol Version 4	96.3	3563	2.6	71372	9141	0	0	0	3563
User Datagram Protocol	49.1	1937	0.6	15496	1984	0	0	0	1937
Simple Service Discovery Protocol	2.9	116	1.2	33778	4326	116	33778	4326	116
QUIC IETF	24.5	965	34.5	955096	122 k	965	951932	121 k	973
NetBIOS Name Service	0.6	24	0.1	1632	209	24	1632	209	24
Multicast Domain Name System	1.9	76	0.8	21682	2800	76	21682	2800	76
Link-local Multicast Name Resolution	0.1	5	0.0	165	21	5	165	21	5
Dynamic Host Configuration Protocol	0.2	6	0.1	1806	231	6	1806	231	6
Dropbox LAN sync Discovery Protocol	0.1	2	0.0	184	23	2	184	23	2
Data	18.8	743	17.2	474927	60 k	743	474927	60 k	743
Transmission Control Protocol	40.3	1590	1.2	32196	4123	1207	24536	3142	1590
Transport Layer Security	8.9	352	38.6	1068294	136 k	352	998621	127 k	361
Hypertext Transfer Protocol	0.1	4	0.0	755	96	2	224	28	4
Line-based text data	0.0	1	0.0	559	71	1	559	71	1
JavaScript Object Notation	0.0	1	0.0	50	6	1	50	6	1
FTP Data	0.0	1	0.0	379	48	0	0	0	1
Line-based text data	0.0	1	0.0	379	48	1	379	48	1
File Transfer Protocol (FTP)	0.6	22	0.0	674	86	22	674	86	22
Data	0.1	4	0.0	4	0	4	4	0	4
Internet Group Management Protocol	0.7	28	0.0	480	61	28	480	61	28
Internet Control Message Protocol	0.2	8	0.0	320	40	8	320	40	8
IEEE 1905.1a	0.1	3	0.0	238	30	3	238	30	3
Address Resolution Protocol	4.3	168	0.2	4704	602	168	4704	602	168

No display filter

Close Copy Protocols Help

tcp.port == 21

tcp.port == 21									
No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info	
549	2026-09-28 06:37:26.971822	192.168.1.50	194.108.117.16	TCP	66			59734 → 21 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
552	2026-09-28 06:37:27.189322	194.108.117.16	192.168.1.50	TCP	66			0.213204700 21 → 59734 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 WS=256 SACK_PERM	
553	2026-09-28 06:37:27.189322	192.168.1.50	194.108.117.16	TCP	54			0.000009700 59734 → 21 [ACK] Seq=1 Ack=1 Win=65535 Len=0	
556	2026-09-28 06:37:27.396689	194.108.117.16	192.168.1.50	FTP	225			0.211675600 Response: 220-Welcome to test.rebex.net!	
557	2026-09-28 06:37:27.397306	192.168.1.50	194.108.117.16	FTP	65			0.000404500 Request: USER demo	
560	2026-09-28 06:37:27.688018	194.108.117.16	192.168.1.50	FTP	130			0.210801200 Response: 331 Anonymous login OK, send your complete email address as your password.	
561	2026-09-28 06:37:27.688028	192.168.1.50	194.108.117.16	FTP	69			0.000105000 Request: PASS password	
562	2026-09-28 06:37:27.818081	194.108.117.16	192.168.1.50	FTP	82			0.210522100 Response: 230 User 'demo' logged in.	
563	2026-09-28 06:37:27.819181	192.168.1.50	194.108.117.16	FTP	59			0.000368200 Request: PWD	
564	2026-09-28 06:37:28.031066	194.108.117.16	192.168.1.50	FTP	85			0.211882400 Response: 257 "/" is current directory.	
565	2026-09-28 06:37:28.031229	192.168.1.50	194.108.117.16	FTP	63			0.000231400 Request: CWD pub	
568	2026-09-28 06:37:28.242122	194.108.117.16	192.168.1.50	FTP	88			0.210630400 Response: 250 Directory changed to "/pub".	
569	2026-09-28 06:37:28.242266	192.168.1.50	194.108.117.16	FTP	67			0.000135900 Request: CWD example	
573	2026-09-28 06:37:28.474002	194.108.117.16	192.168.1.50	FTP	96			0.231760900 Response: 250 Directory changed to "/pub/example".	
574	2026-09-28 06:37:28.474444	192.168.1.50	194.108.117.16	FTP	60			0.000421600 Request: EPSV	
576	2026-09-28 06:37:28.687356	194.108.117.16	192.168.1.50	FTP	101			0.212910700 Response: 229 Entering Extended Passive Mode (1041)	
578	2026-09-28 06:37:28.739044	192.168.1.50	194.108.117.16	TCP	54			0.051692100 59734 → 21 [ACK] Seq=60 Ack=430 Win=65024 Len=0	
583	2026-09-28 06:37:28.925676	192.168.1.50	194.108.117.16	FTP	62			0.186626100 Request: TYPE I	
584	2026-09-28 06:37:29.135690	194.108.117.16	192.168.1.50	FTP	74			0.210225000 Response: 200 TYPE set to I.	
585	2026-09-28 06:37:29.136256	192.168.1.50	194.108.117.16	FTP	71			0.000358200 Request: SIZE readme.txt	
589	2026-09-28 06:37:29.347576	194.108.117.16	192.168.1.50	FTP	63			0.211315000 Response: 213 379	
590	2026-09-28 06:37:29.348004	192.168.1.50	194.108.117.16	FTP	71			0.000477600 Request: RETR readme.txt	
598	2026-09-28 06:37:29.569085	194.108.117.16	192.168.1.50	FTP	117			0.221808500 Response: 125 Data connection already open; starting "BINARY" transfer.	
600	2026-09-28 06:37:29.611886	192.168.1.50	194.108.117.16	TCP	54			0.042031900 59734 → 21 [ACK] Seq=102 Ack=522 Win=64768 Len=0	
603	2026-09-28 06:37:29.787713	194.108.117.16	192.168.1.50	FTP	78			0.175250200 Response: 226 Transfer complete.	
604	2026-09-28 06:37:29.792266	192.168.1.50	194.108.117.16	FTP	60			0.005868100 Request: QUIT	
607	2026-09-28 06:37:30.003441	194.108.117.16	192.168.1.50	FTP	76			0.211205900 Response: 221 Closing session.	
608	2026-09-28 06:37:30.003441	194.108.117.16	192.168.1.50	TCP	54			0.000000000 21 → 59734 [FIN, ACK] Seq=568 Ack=108 Win=2096896 Len=0	
609	2026-09-28 06:37:30.003449	192.168.1.50	194.108.117.16	TCP	54			0.000082500 59734 → 21 [ACK] Seq=108 Ack=569 Win=64768 Len=0	
610	2026-09-28 06:37:30.003592	192.168.1.50	194.108.117.16	TCP	54			0.000027300 59734 → 21 [FIN, ACK] Seq=108 Ack=569 Win=64768 Len=0	
611	2026-09-28 06:37:30.223546	194.108.117.16	192.168.1.50	TCP	54			0.220026500 21 → 59734 [ACK] Seq=569 Ack=109 Win=2096896 Len=0	
612	2026-09-28 06:37:30.223546	194.108.117.16	192.168.1.50	TCP	54			0.000000000 [TCP_Zerowindow] 21 → 59734 [ACK] Seq=569 Ack=109 Win=0 Len=0	

Wireshark - Protocol Hierarchy Statistics - capstone.pcapng									
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
▼ Frame	100.0	32	100.0	2426	5968	0	0	0	32
▼ Ethernet	100.0	32	18.5	448	1102	0	0	0	32
▼ Internet Protocol Version 4	100.0	32	26.4	640	1574	0	0	0	32
▼ Transmission Control Protocol	100.0	32	27.4	664	1633	10	224	551	32
File Transfer Protocol (FTP)	68.8	22	27.8	674	1658	22	674	1658	22

Display filter: tcp.port == 21

Close Copy Protocols Help

ip.addr == 194.108.117.16

capstone.pcapng									
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help									
ip.addr == 194.108.117.16									
No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info	
549	2026-09-28 06:37:26.971822	192.168.1.50	194.108.117.16	TCP	66			59734 → 21 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
552	2026-09-28 06:37:27.189322	194.108.117.16	192.168.1.50	TCP	66			0.213204700 21 → 59734 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 WS=256 SACK_PERM	
553	2026-09-28 06:37:27.189322	192.168.1.50	194.108.117.16	TCP	54			0.000097700 59734 → 21 [ACK] Seq=1 Ack=1 Win=65200 Len=0	
556	2026-09-28 06:37:27.396680	194.108.117.16	192.168.1.50	FTP	225			0.211675600 Response: 220-Welcome to test.rebex.net!	
557	2026-09-28 06:37:27.397380	192.168.1.50	194.108.117.16	FTP	65			0.000404500 Request: USER demo	
560	2026-09-28 06:37:27.608018	194.108.117.16	192.168.1.50	FTP	130			0.210801200 Response: 331 Anonymous login OK, send your complete email address as your password.	
561	2026-09-28 06:37:27.608020	192.168.1.50	194.108.117.16	FTP	69			0.000105000 Request: PASS password	
562	2026-09-28 06:37:27.818081	194.108.117.16	192.168.1.50	FTP	82			0.210522100 Response: 230 User 'demo' logged in.	
563	2026-09-28 06:37:27.819181	192.168.1.50	194.108.117.16	FTP	59			0.000368200 Request: PWD	
564	2026-09-28 06:37:28.031086	194.108.117.16	192.168.1.50	FTP	85			0.211882400 Response: 257 "/" is current directory.	
565	2026-09-28 06:37:28.031229	192.168.1.50	194.108.117.16	FTP	63			0.000231400 Request: CWD pub	
568	2026-09-28 06:37:28.242121	194.108.117.16	192.168.1.50	FTP	88			0.210830400 Response: 250 Directory changed to "/pub".	
569	2026-09-28 06:37:28.242226	192.168.1.50	194.108.117.16	FTP	67			0.000135900 Request: CWD example	
573	2026-09-28 06:37:28.474002	194.108.117.16	192.168.1.50	FTP	96			0.231760900 Response: 250 Directory changed to "/pub/example".	
574	2026-09-28 06:37:28.474444	192.168.1.50	194.108.117.16	FTP	68			0.000421600 Request: QPVS	
576	2026-09-28 06:37:28.687351	194.108.117.16	192.168.1.50	FTP	101			0.212910700 Response: 229 Entering Extended Passive Mode (1041)	
577	2026-09-28 06:37:28.687611	192.168.1.50	194.108.117.16	TCP	66			0.000261400 59738 → 1041 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	
578	2026-09-28 06:37:28.739041	192.168.1.50	194.108.117.16	TCP	54			0.051430700 59734 → 21 [ACK] Seq=60 Ack=430 Win=65824 Len=0	
581	2026-09-28 06:37:28.904806	194.108.117.16	192.168.1.50	TCP	66			0.165017200 1041 → 59738 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 WS=256 SACK_PERM	
582	2026-09-28 06:37:28.904825	192.168.1.50	194.108.117.16	TCP	54			0.000188400 59738 → 1041 [ACK] Seq=1 Ack=1 Win=65200 Len=0	
583	2026-09-28 06:37:28.925671	192.168.1.50	194.108.117.16	FTP	62			0.021420500 Request: TYPE I	
584	2026-09-28 06:37:29.135589	194.108.117.16	192.168.1.50	FTP	74			0.210226000 Response: 200 TYPE set to I.	
585	2026-09-28 06:37:29.136225	192.168.1.50	194.108.117.16	FTP	71			0.000355200 Request: SIZE readme.txt	
589	2026-09-28 06:37:29.347571	194.108.117.16	192.168.1.50	FTP	63			0.211315000 Response: 213 379	
590	2026-09-28 06:37:29.348084	192.168.1.50	194.108.117.16	FTP	71			0.000477600 Request: RETR readme.txt	
594	2026-09-28 06:37:29.559596	194.108.117.16	192.168.1.50	FTP-DATA	433			0.211451500 FTP Data: 379 bytes (EPASV) (RETR readme.txt)	
595	2026-09-28 06:37:29.559596	194.108.117.16	192.168.1.50	TCP	54			0.000000000 1041 → 59738 [FIN, ACK] Seq=908 Ack=1 Win=0 Seq=97152 Len=0	
596	2026-09-28 06:37:29.559606	192.168.1.50	194.108.117.16	TCP	54			0.000194300 59738 → 1041 [ACK] Seq=1 Ack=381 Win=65024 Len=0	
598	2026-09-28 06:37:29.560885	194.108.117.16	192.168.1.50	FTP	117			0.010162700 Response: 125 Data connection already open; starting 'BINARY' transfer.	
599	2026-09-28 06:37:29.570592	192.168.1.50	194.108.117.16	TCP	54			0.000666400 59738 → 1041 [FIN, ACK] Seq=1 Ack=381 Win=65024 Len=0	
600	2026-09-28 06:37:29.611881	192.168.1.50	194.108.117.16	TCP	54			0.035367700 59734 → 21 [ACK] Seq=102 Ack=522 Win=64768 Len=0	
602	2026-09-28 06:37:29.708017	194.108.117.16	192.168.1.50	TCP	54			0.168289000 1041 → 59738 [ACK] Seq=381 Ack=2 Win=2097152 Len=0	
603	2026-09-28 06:37:29.707713	194.108.117.16	192.168.1.50	FTP	78			0.000061200 Response: 226 Transfer complete.	
604	2026-09-28 06:37:29.792226	192.168.1.50	194.108.117.16	FTP	68			0.005068100 Request: QUIT	
607	2026-09-28 06:37:30.003411	194.108.117.16	192.168.1.50	FTP	76			0.211205000 Response: 221 Closing session.	
608	2026-09-28 06:37:30.003411	194.108.117.16	192.168.1.50	TCP	54			0.000000000 21 → 59734 [FIN, ACK] Seq=568 Ack=108 Win=2096896 Len=0	
609	2026-09-28 06:37:30.003491	192.168.1.50	194.108.117.16	TCP	54			0.000082500 59734 → 21 [ACK] Seq=108 Ack=569 Win=64768 Len=0	
610	2026-09-28 06:37:30.003592	192.168.1.50	194.108.117.16	TCP	54			0.000027000 59734 → 21 [FIN, ACK] Seq=108 Ack=569 Win=64768 Len=0	
611	2026-09-28 06:37:30.223541	194.108.117.16	192.168.1.50	TCP	54			0.220025500 21 → 59734 [ACK] Seq=569 Ack=109 Win=2096896 Len=0	
612	2026-09-28 06:37:30.223541	194.108.117.16	192.168.1.50	TCP	54			0.000000000 [TCP ZeroWindow] 21 → 59734 [ACK] Seq=569 Ack=109 Win=0 Len=0	

frame contains "labuser"

capstone.pcapng									
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help									
frame contains 'labuser'									
No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info	
678	2026-09-28 06:37:32.596338	192.168.1.50	34.200.179.76	HTTP	203			GET /basic-auth/labuser/Lab12345 HTTP/1.1	
701	2026-09-28 06:37:32.829222	34.200.179.76	192.168.1.50	HTTP/3..	184			0.232842200 HTTP/1.1 200 OK , JSON (application/json)	

File Edit View Go Capture Analysis Statistics Tools Help

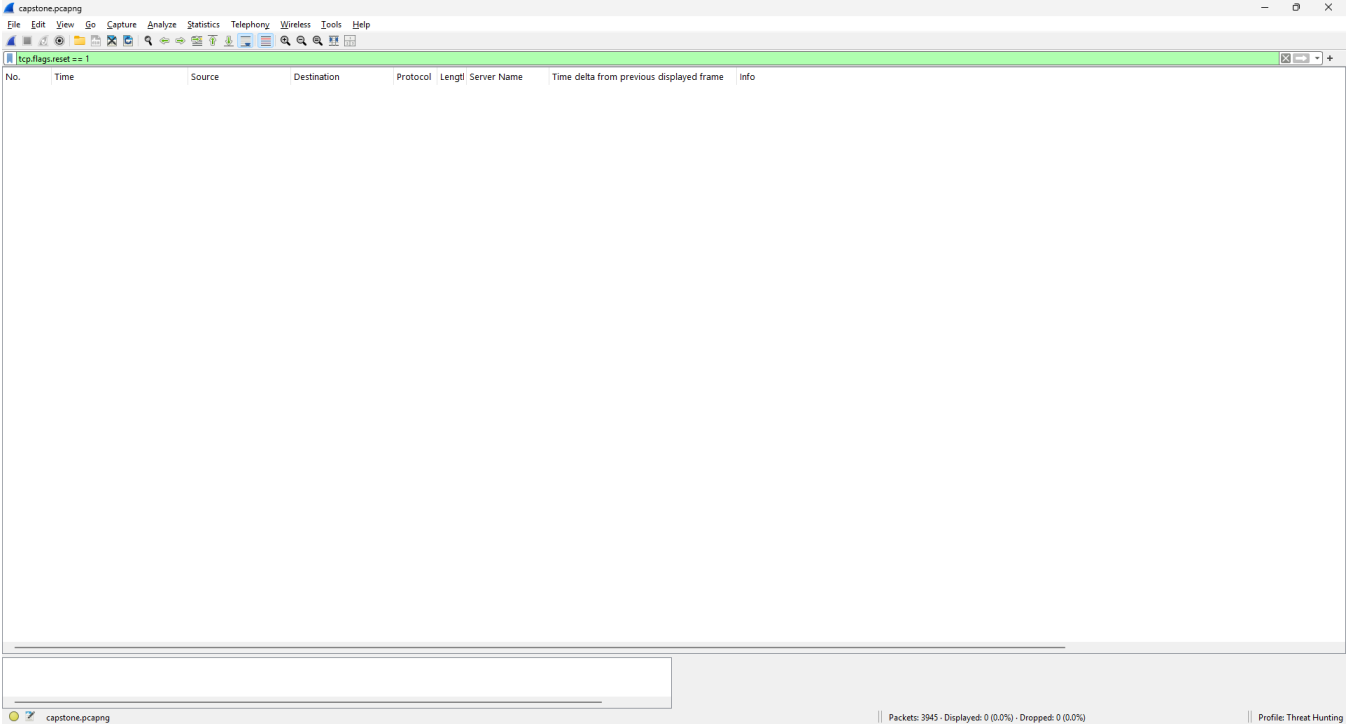
ip.src == 192.168.1.50 & http.port == 80 || tcp.port == 21

192.168.1.50

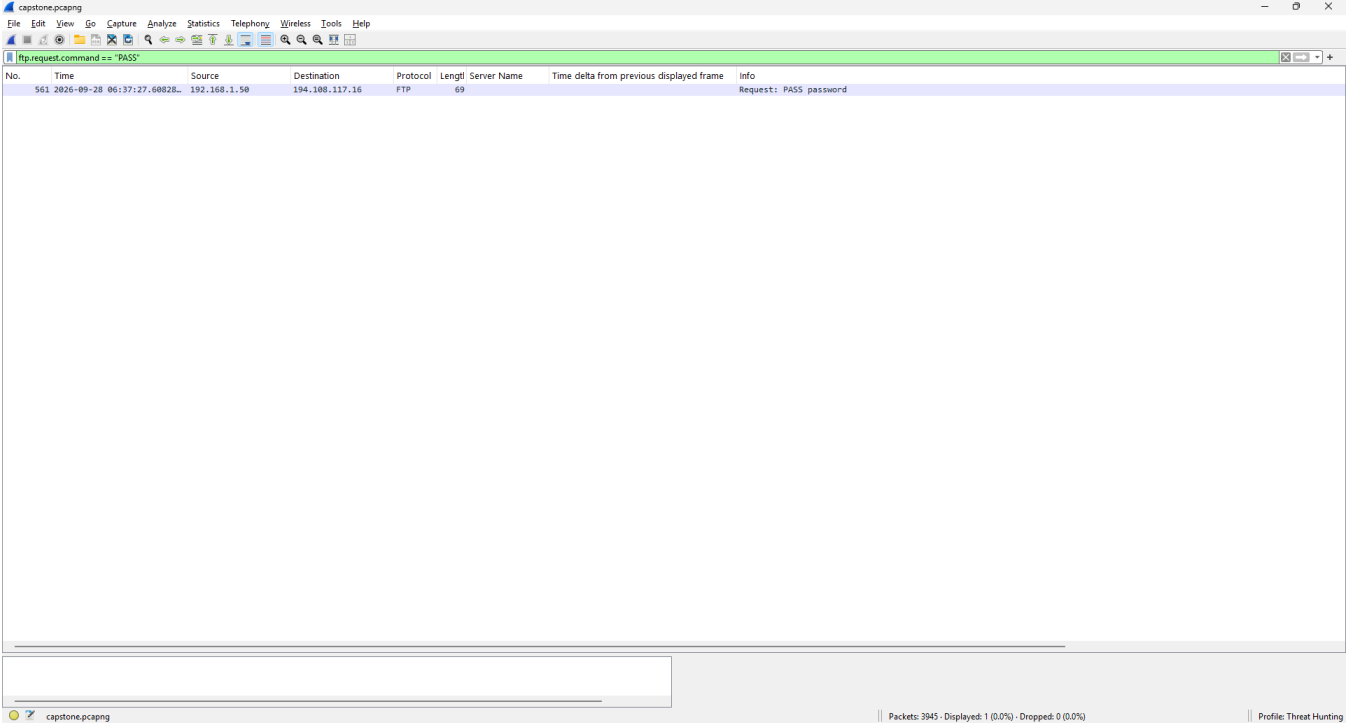
80 || tcp.port == 21

No.	Time	Source	Destination	Protocol	Length	Server Name	Time delta from previous displayed frame	Info
461	2026-09-28 06:37:23.30853..	192.168.1.50	172.66.147.243	TCP	66			59714 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
463	2026-09-28 06:37:23.32284..	192.168.1.50	172.66.147.243	TCP	54			0.014316900 59714 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
464	2026-09-28 06:37:23.32298..	192.168.1.50	172.66.147.243	HTTP	129			0.000131000 GET / HTTP/1.1
468	2026-09-28 06:37:23.34536..	192.168.1.50	172.66.147.243	TCP	54			0.022387600 59714 → 80 [ACK] Seq=76 Ack=874 Win=64512 Len=0
469	2026-09-28 06:37:23.34644..	192.168.1.50	172.66.147.243	TCP	54			0.001076600 59714 → 80 [FIN, ACK] Seq=76 Ack=874 Win=64512 Len=0
471	2026-09-28 06:37:23.36239..	192.168.1.50	172.66.147.243	TCP	54			0.015951400 59714 → 80 [ACK] Seq=77 Ack=875 Win=64512 Len=0
549	2026-09-28 06:37:26.97162..	192.168.1.50	194.108.117.16	TCP	66			3.688482300 59734 → 21 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
553	2026-09-28 06:37:27.18522..	192.168.1.50	194.108.117.16	TCP	54			0.213392400 59734 → 21 [ACK] Seq=1 Ack=1 Win=65280 Len=0
557	2026-09-28 06:37:27.39730..	192.168.1.50	194.108.117.16	FTP	65			0.120889300 Request: USER demo
561	2026-09-28 06:37:27.68828..	192.168.1.50	194.108.117.16	FTP	69			0.210989300 Request: PASS password
563	2026-09-28 06:37:27.81918..	192.168.1.50	194.108.117.16	FTP	59			0.210890300 Request: PWD
565	2026-09-28 06:37:28.88129..	192.168.1.50	194.108.117.16	FTP	63			0.212113300 Request: CWD pub
569	2026-09-28 06:37:28.24226..	192.168.1.50	194.108.117.16	FTP	67			0.210966300 Request: CWD example
574	2026-09-28 06:37:28.47444..	192.168.1.50	194.108.117.16	FTP	60			0.232182500 Request: EPSV
578	2026-09-28 06:37:28.70904..	192.168.1.50	194.108.117.16	TCP	54			0.264602000 59734 → 21 [ACK] Seq=60 Ack=430 Win=65824 Len=0
583	2026-09-28 06:37:28.92567..	192.168.1.50	194.108.117.16	FTP	62			0.186262100 Request: TYPE I
585	2026-09-28 06:37:29.13625..	192.168.1.50	194.108.117.16	FTP	71			0.210584200 Request: SIZE reader.txt
590	2026-09-28 06:37:29.34804..	192.168.1.50	194.108.117.16	FTP	60			0.211792600 Request: RETR reader.txt
600	2026-09-28 06:37:29.61188..	192.168.1.50	194.108.117.16	TCP	54			0.263848400 59734 → 21 [ACK] Seq=182 Ack=522 Win=64768 Len=0
604	2026-09-28 06:37:29.79220..	192.168.1.50	194.108.117.16	FTP	67			0.180131300 Request: QUIT
609	2026-09-28 06:37:30.08349..	192.168.1.50	194.108.117.16	TCP	54			0.211288400 59734 → 21 [ACK] Seq=188 Ack=569 Win=64768 Len=0
618	2026-09-28 06:37:30.08352..	192.168.1.50	194.108.117.16	TCP	54			0.000027300 59734 → 21 [FIN, ACK] Seq=188 Ack=569 Win=64768 Len=0
662	2026-09-28 06:37:32.35827..	192.168.1.50	34.200.179.76	TCP	66			2.354754000 59754 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
677	2026-09-28 06:37:32.59622..	192.168.1.50	34.200.179.76	TCP	54			0.237947000 59754 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
678	2026-09-28 06:37:32.59638..	192.168.1.50	34.200.179.76	HTTP	203			0.000161900 GET /basic-auth/labuser/lab12345 HTTP/1.1
702	2026-09-28 06:37:32.82941..	192.168.1.50	34.200.179.76	TCP	54			0.233025800 59754 → 80 [ACK] Seq=150 Ack=280 Win=65824 Len=0
703	2026-09-28 06:37:32.83039..	192.168.1.50	34.200.179.76	TCP	54			0.000094800 59754 → 80 [FIN, ACK] Seq=150 Ack=280 Win=65824 Len=0
707	2026-09-28 06:37:33.06371..	192.168.1.50	34.200.179.76	TCP	54			0.233316

tcp.flags.reset == 1



ftp.request.command == "PASS"



Screenshot	Use it?	Note
Unfiltered Protocol Hierarchy (3,945 packets)	Yes	Protocols present

tcp.port == 21 (32 packets)	Yes	USER demo (frame 557) and PASS password (frame 561) are visible in the Info column
Frame 678, first view	No	The HTTP section is collapsed, so no credentials show
Frame 678, scrolled down	Yes, but retake	It shows Credentials: labuser:Lab12345, but a Snipping Tool notification covers the bottom-right corner. Wait for it to disappear and retake, or crop it out.
frame contains "labuser" (2 packets)	Yes	Filter 1 (frames 678 and 701)
ip.src == 192.168.1.50 && (tcp.port == 80 tcp.port == 21) (28 packets)	Yes	Filter 2, showing all three cleartext destinations
tcp.flags.reset == 1 (0 packets)	Yes	Filter 3, no forced resets

Incident Report: Cleartext Credential Exposure (capstone.pcapng)

Protocols present. The capture has 3,945 packets (2.76 MB). By bytes it is mostly encrypted: TLS 38.6% and QUIC 34.5%. Cleartext application traffic is small: 4 HTTP packets and 22 FTP packets. Local discovery traffic (SSDP, mDNS, NetBIOS, ARP) is also present.

Troubleshooting Quick Reference

Problem	What to try
Browser keeps switching http:// to https://	Use <code>curl</code> (Method 1) or turn off your browser's HTTPS-first setting (Method 2). Use <code>example.com</code> if <code>neverssl.com</code> is blocked.
No DNS packets appear	Your browser may use encrypted DNS. Run <code>nslookup example.com</code> in Command Prompt and filter on <code>dns</code> .
<code>curl</code> output looks strange or the command fails	Use Command Prompt, or type <code>curl.exe</code> in PowerShell.
FTP connection times out	Port 21 is probably blocked on your network. Use Path A or Path C.
FTP login works but <code>dir</code> hangs	Expected with the Windows ftp client. Type <code>bye</code> ; the credentials were already captured. Or use <code>curl ftp://demo:password@test.rebex.net/</code> .
<code>httpbin.org</code> is down or errors	Try <code>httpbingo.org</code> with the same <code>/basic-auth/user/pass</code> path, or use Path B or C.
<code>sftp</code> is not recognized	Enable "OpenSSH Client" under Settings → Apps → Optional features (needed only for the extra-credit step).
Wireshark shows no interfaces or cannot capture	Run Wireshark as Administrator and confirm Npcap is installed.
Too much unrelated traffic in the capture	Close other apps and browser tabs, use a capture filter (Activity 2), or narrow with a display filter such as <code>ip.addr == <server IP></code> .